

Finite Element Method- Programs and Applications

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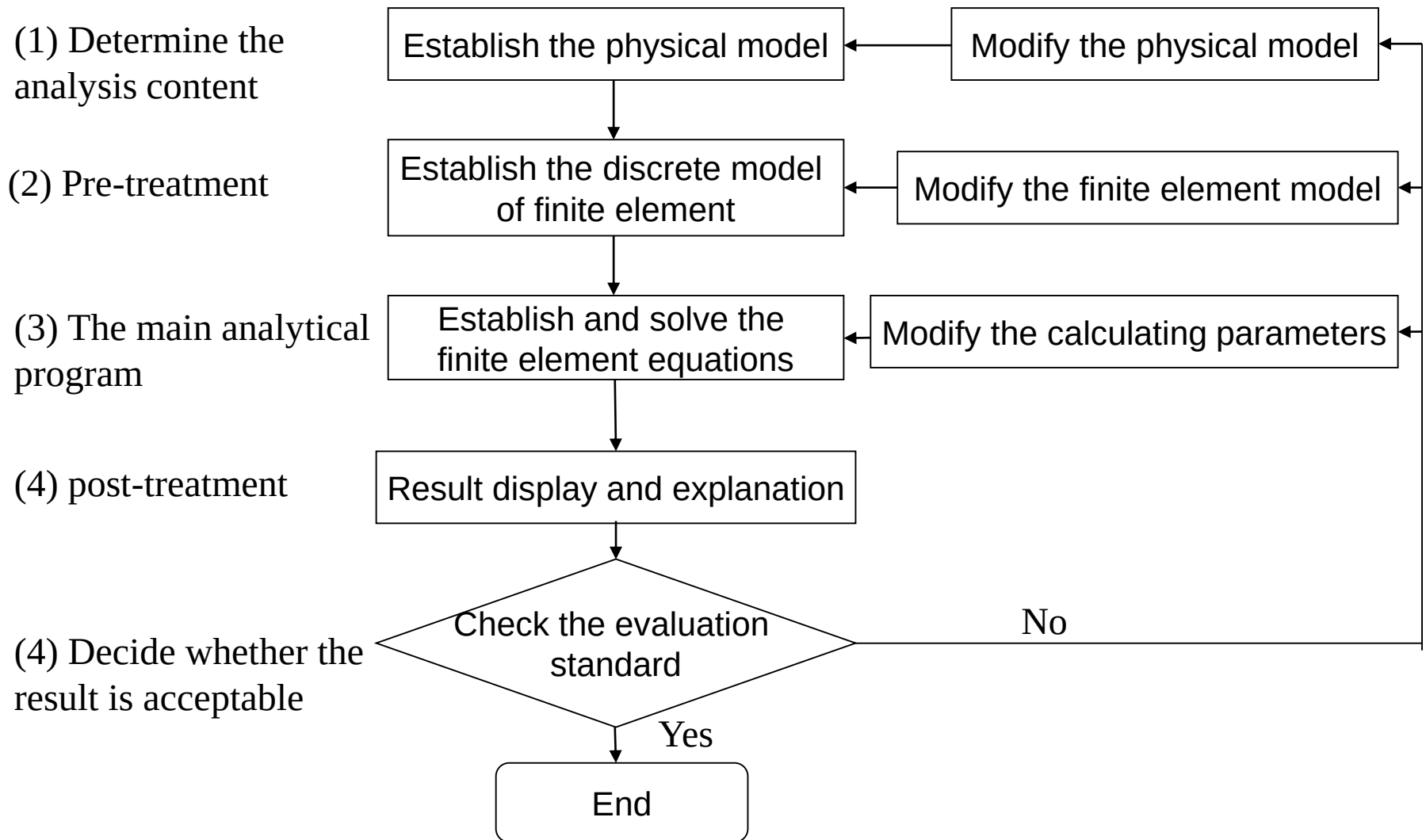
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1. Programs

- ABAQUS (SUBROUTINE)
- ANSYS (LS-DYNA)
- SAP2000 (ETABS, SAFE, PERFORM3D)
- *MSC (PATRA, NASTRAN)*
- *MIDAS (CIVIL, BRIDGE)*
- *LUSAS, etc.*

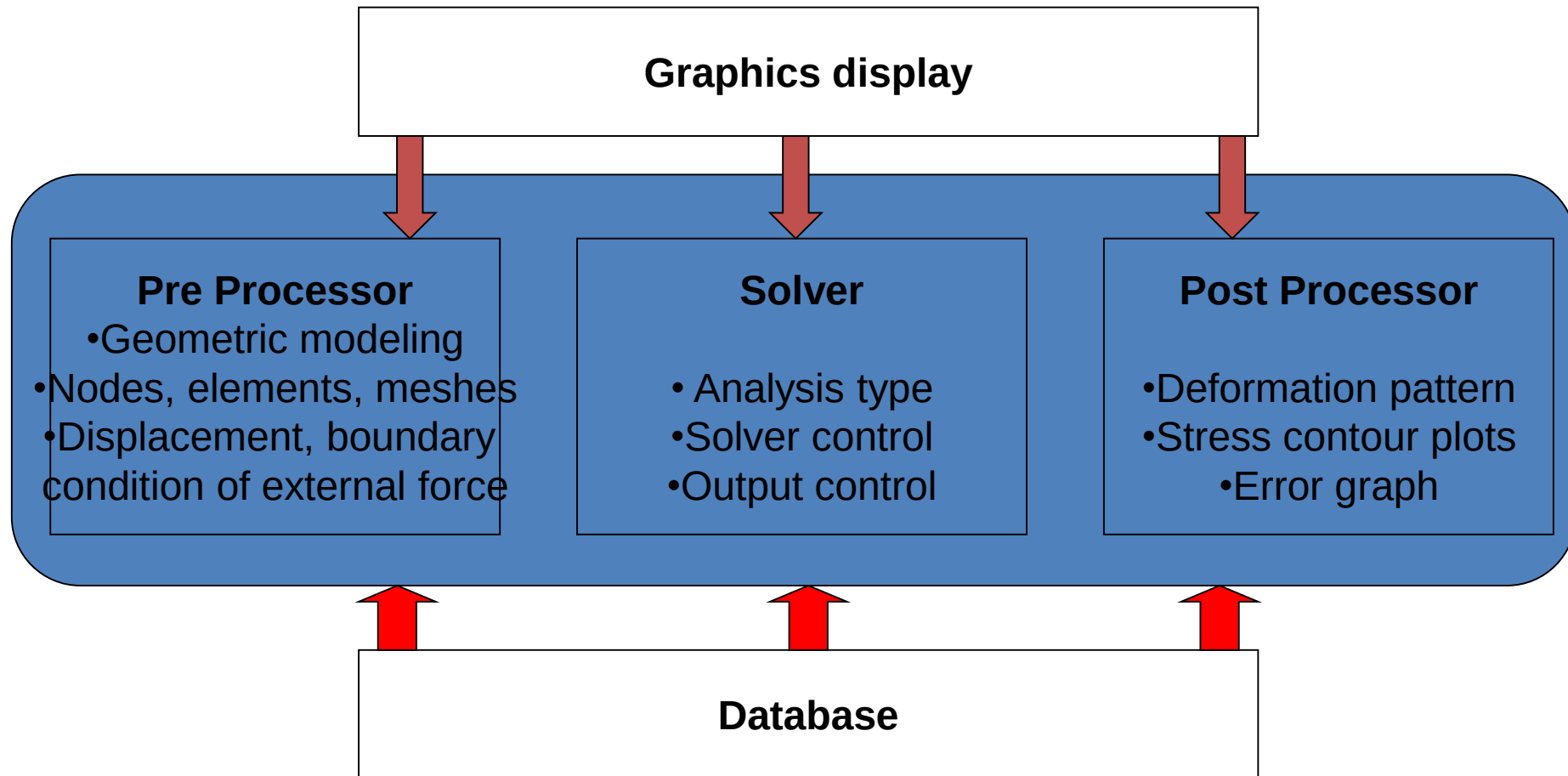
1.1 The analysis flow

The finite element analysis of general engineering and scientific problems can be concluded as follows.

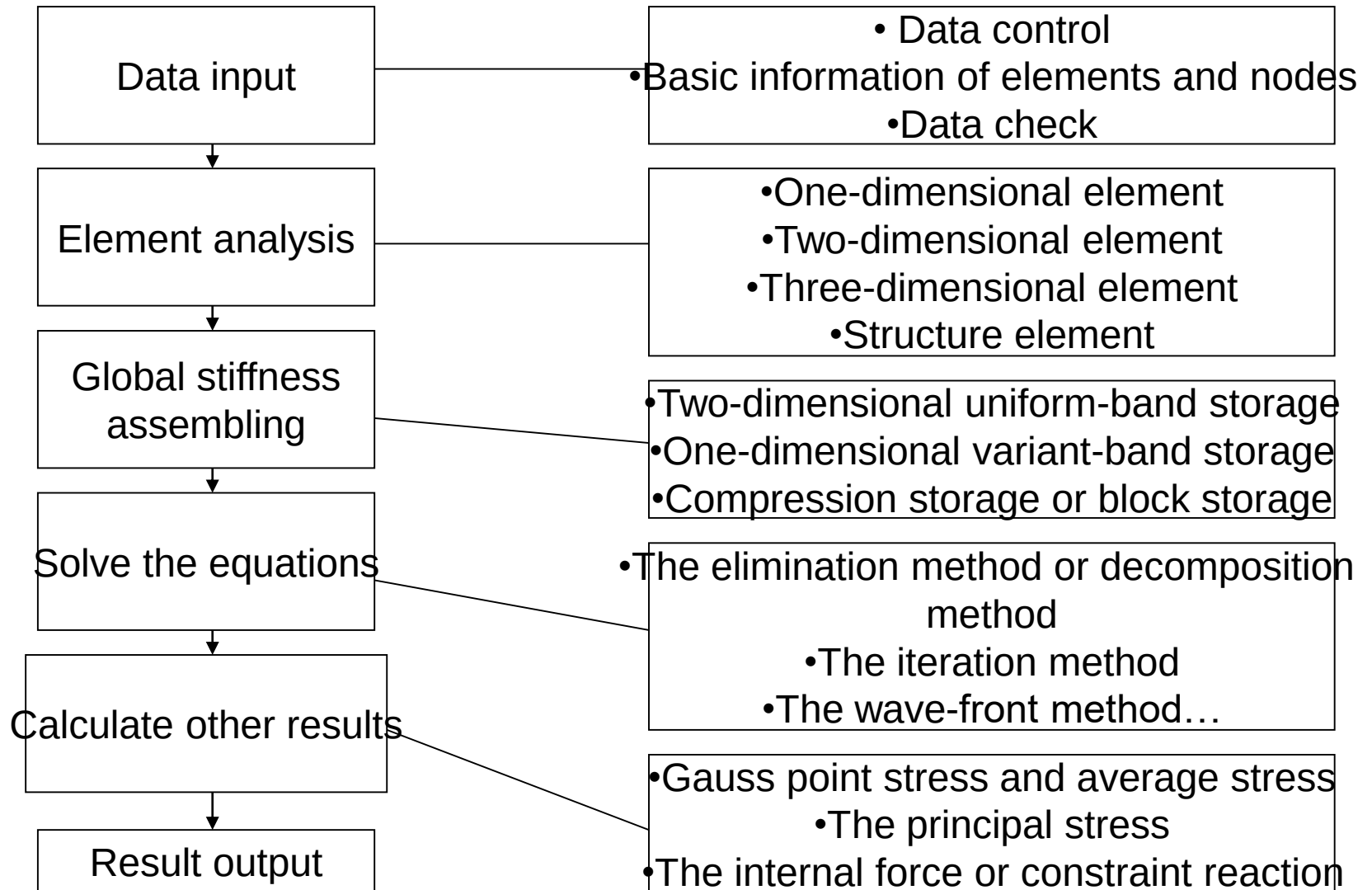


1.2 The analysis platform of finite element program

Corresponding to the analysis procedure mentioned before a complete software of finite element analysis includes three parts and two supporting environments.



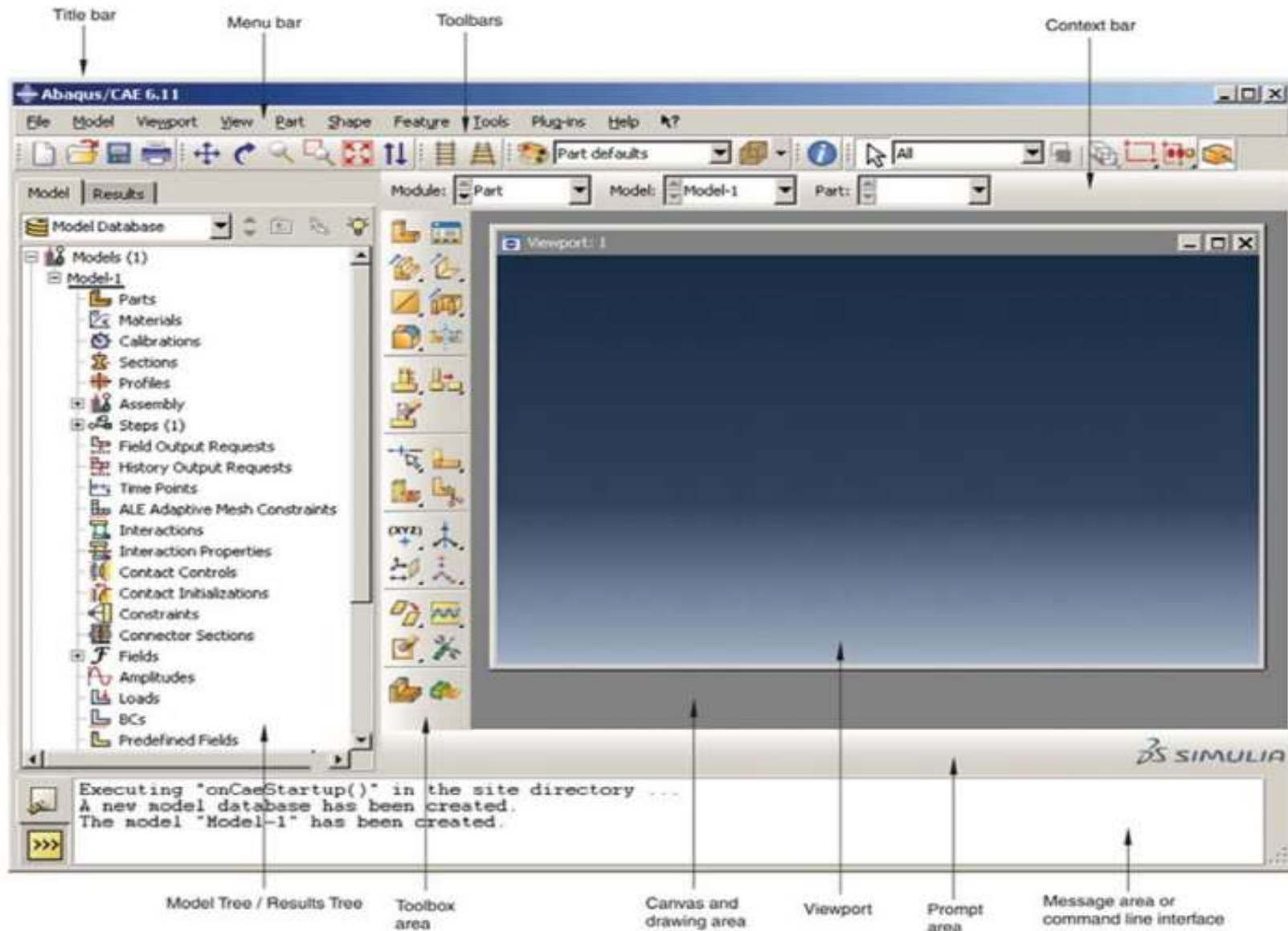
1.3 Critical design procedures



1.4 Units

	SI		SI(mm)		US Unit
Length	m		mm		in
Force	N		N		1bf
Mass	kg		T (10^3kg)		1bf s ² /in
Time	s	↔	s	↔	s
Stress	Pa		MPa		psi
Energy	J		mJ (10^{-3}J)		in 1bf
Density	Kg/m ³		t/mm ³		1bf s ² /in ⁴
Ace	m/s ²		mm/s ²		

ABAQUS



Procedure

1、几何建模 Part

2、特性设置 Property

3、建立装配体 Assembly

4、定义分析步 Step

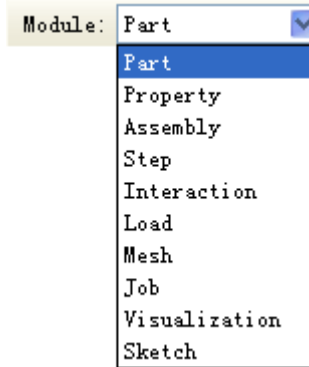
5、相互作用 Interaction

6、载荷边界 Load & BC

7、划分网格 Mesh

8、提交运算 Job

9、后处理 Visualization



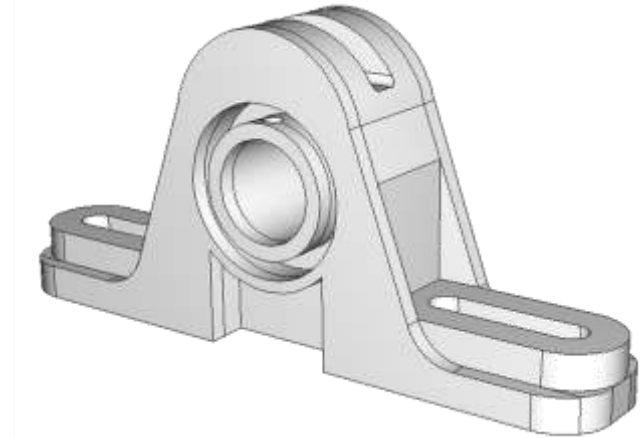
1. Part

– Deformable

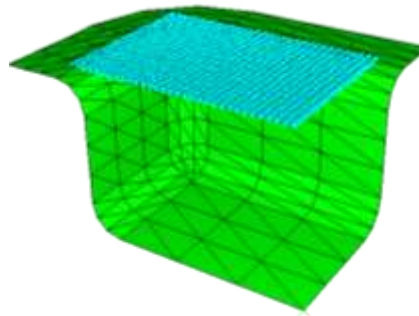
- To create a part with the shape can be changed during the analysis;

– Discrete (Analytical) rigid

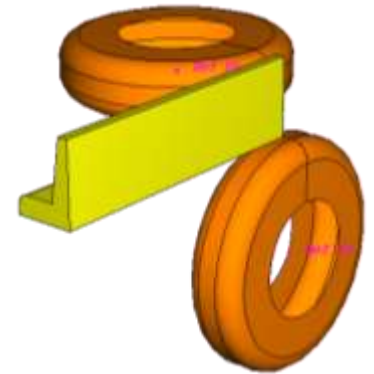
- To create a part with significant stiffness and can not be changed geometrically during the analysis.



可变形部件: 轴机架



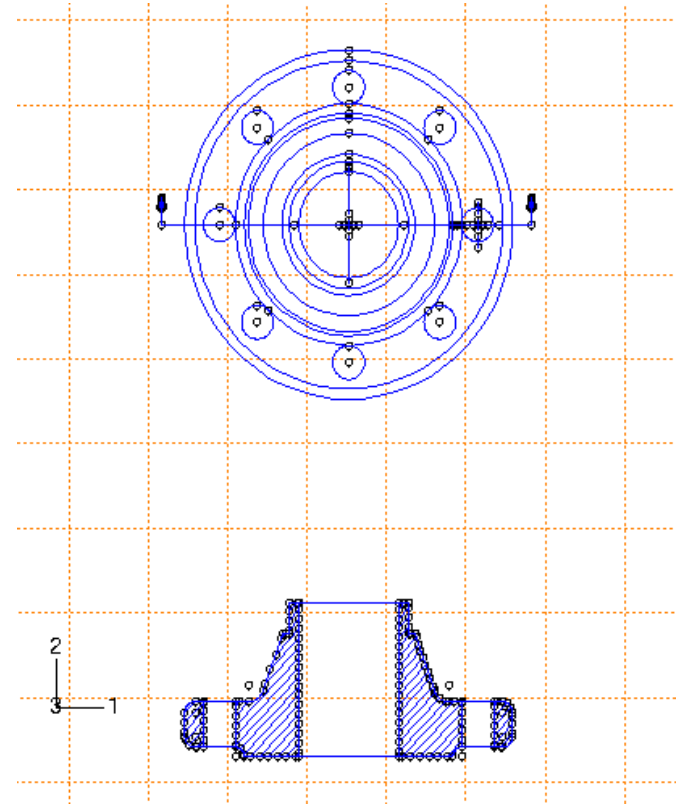
离散刚体部件: 任意形状的刚体冲垫 (带有可变形的毛坯)



解析刚体部件: 旋转的刚性轧滚 (带有可变形的金属原料)

Import geometric model

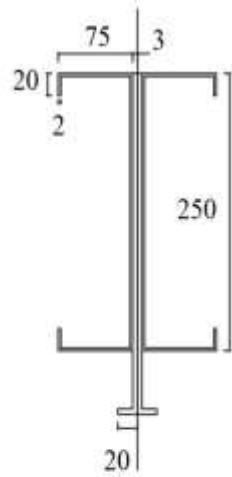
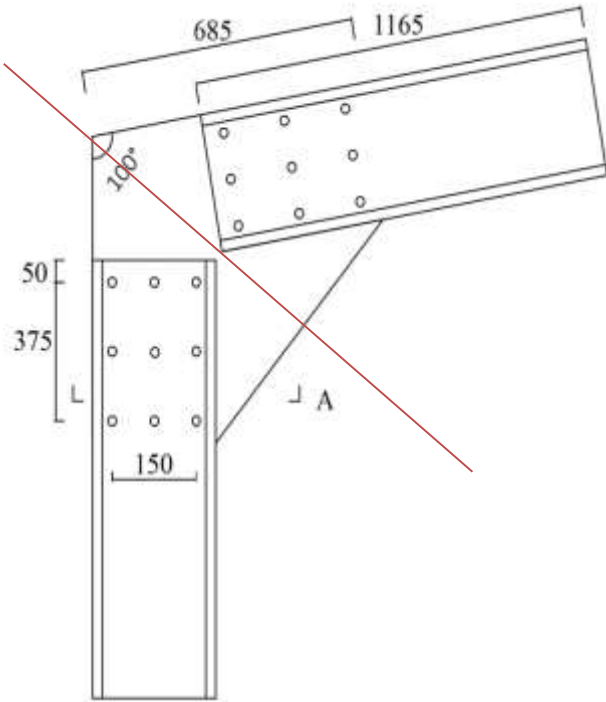
- Skech



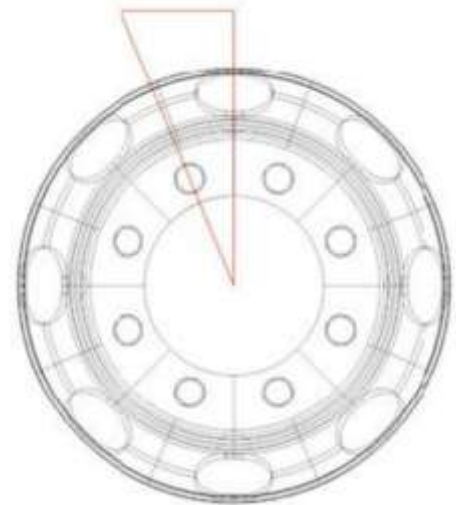
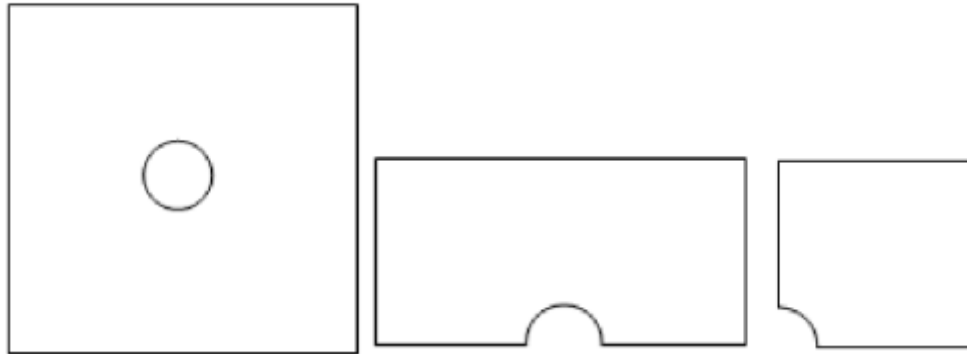
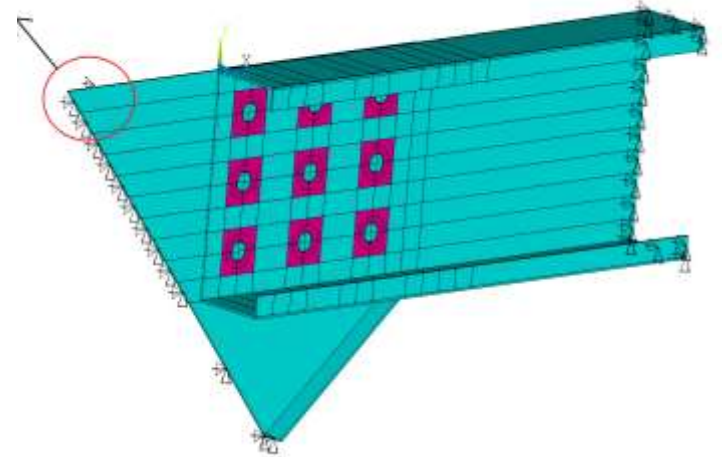
- import

- AutoCAD (.dxf)
- IGES (.igs)
- STEP (.stp)
- ACIS (.sat)

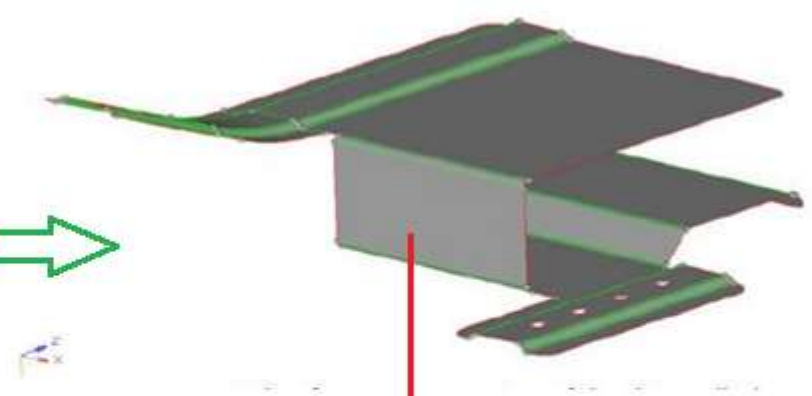
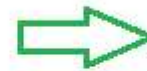
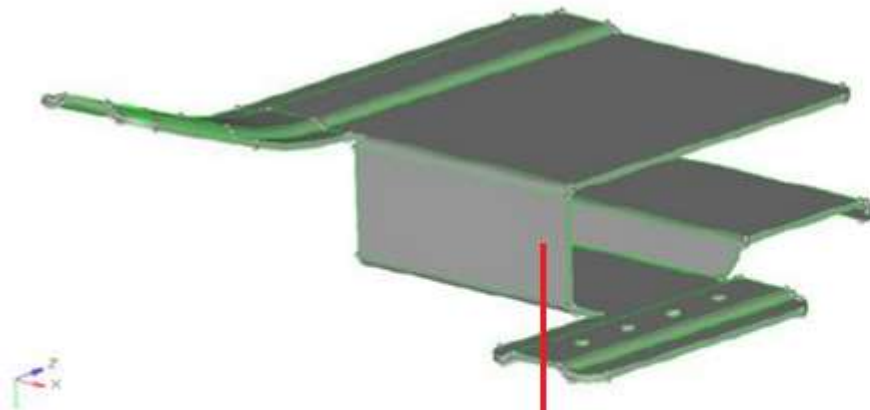
Symmetrical model



剖面A-A



Simplified model



Solid element

Shell element

2 Properties

1) Elasticity →

Elastic (弹性)

Hyperelastic (超弹性)

Hyperfoam (弹性泡沫)

Hypoelastic (亚弹性)

Porous Elastic (多孔弹性)

Viscoelastic (粘弹性)

2) Plasticity

3) Damage for Ductile Metals

4) Damage for Traction Separation (cohesive)

5) Damage for Fiber-Reinforced Composites

6) Deformation Plasticity

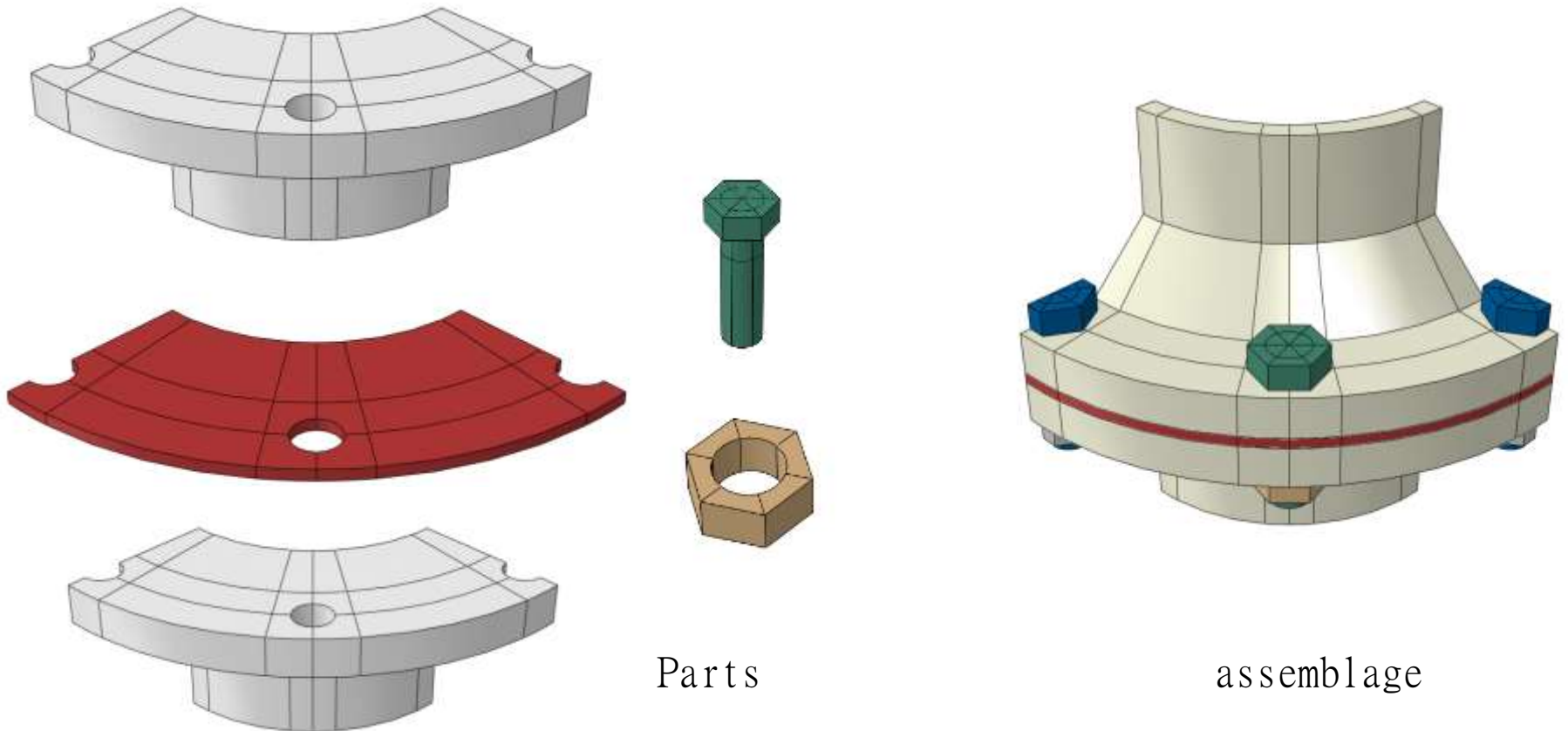
7) Damping

8) Expansion

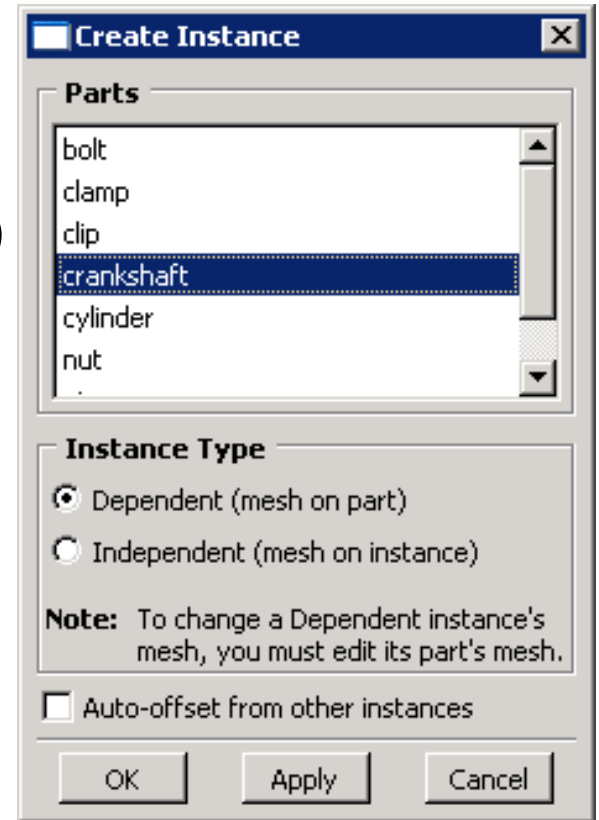
9) Brittle Cracking

3 Assembly

- The assembly module, can be used to position the parts relative to each other in a global coordinate system and to establish a instance.



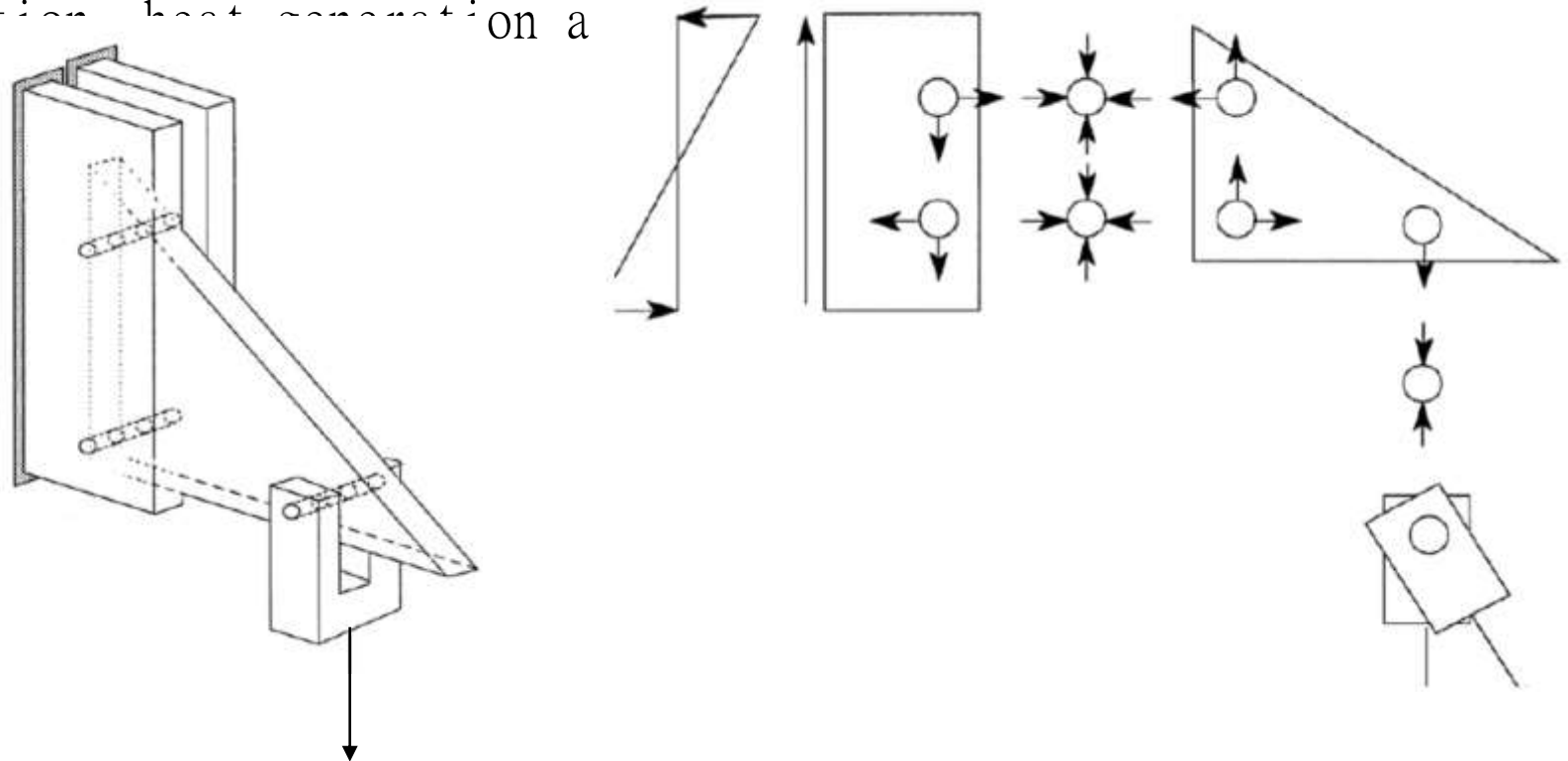
- Dependent and Independent
 - Dependent (mesh on part)
 - Use to mesh the part.
 - Independent (mesh on instance)
 - Use to mesh the instance.



4 Interaction

In the Interaction module you specify mechanical and thermal interactions between regions of a model or between a region of a model and its surroundings. An example of an interaction is contact between two surfaces.

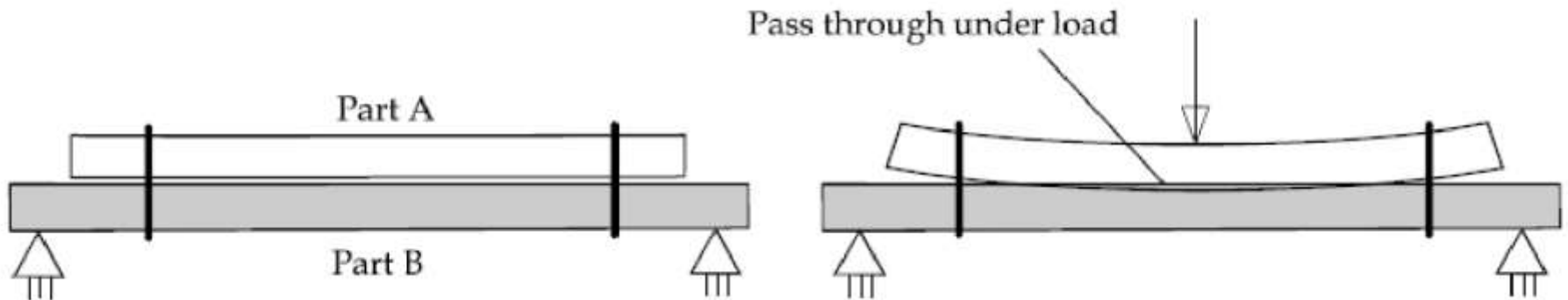
Surface-to-surface Contact needs a definition of master and slave surface, the contact behaviour including friction, damping, heat convection, heat generation on a



The constraint in ABAQUS including:

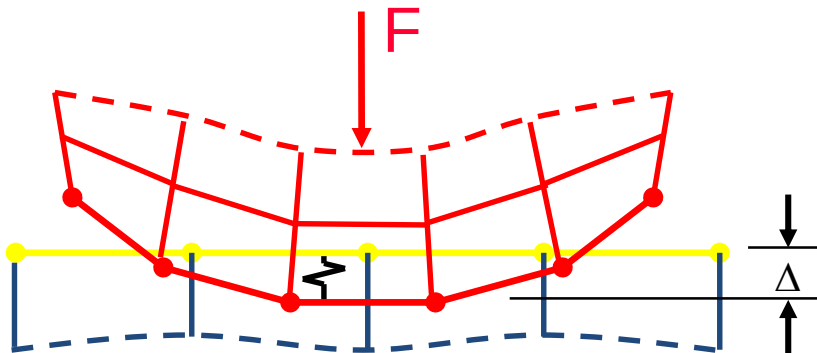
- tie (绑定约束)
 - rigid body (刚体约束)
 - display body (显示体约束)
 - coupling (耦合约束)
 - shell-to-solid coupling (壳-实体耦合约束)
 - embedded region (嵌入约束)
 - equation (约束方程)
-
- **Tie** : is used to bond two regions together, and to disallow the relative movement, the tie contacted regions could have different mesh size.
 - **Rigid body** : is used to create a rigid region in which the relative position of nodes and elements is fixed.
 - **Coupling**: is used to constrain a surface with a reference point.

- Many engineering problems involve contact between two or more components. In these problems a force normal to the contacting surfaces acts on the two bodies when they touch each other. If there is friction between the surfaces, shear forces may be created that resist the tangential motion (sliding) of the bodies.
- The general aim of contact is to:
 - *To resist penetration;*
 - *To transmit normal and tangential friction;*
 - *To ensure free-to-separation.*



penalty method

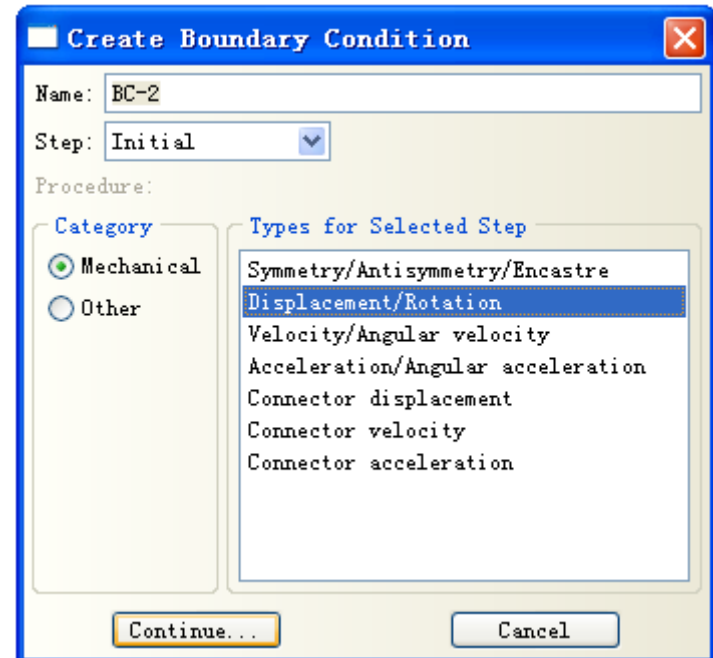
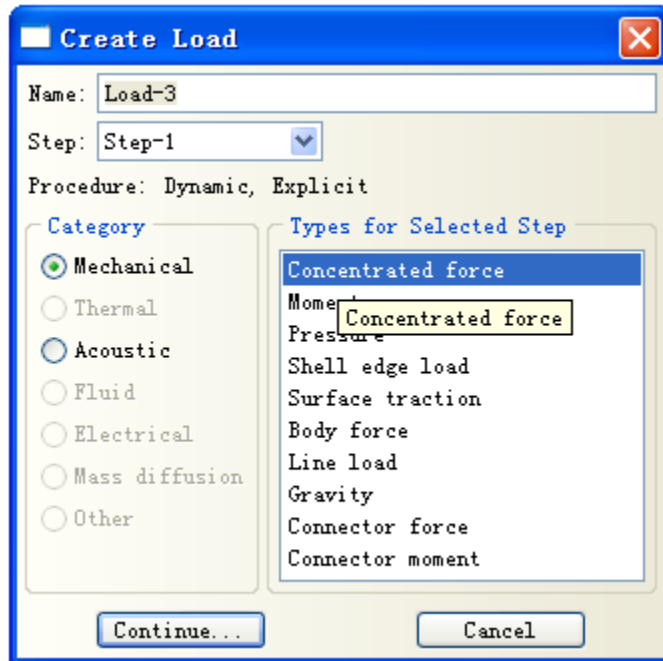
- The method used a contact spring to replace the interaction between surfaces.
 - The spring stiffness is the penalty parameter (contact stiffness).
 - The spring is in deactivate when surfaces are separated, and the spring is active when surfaces are penetrated.



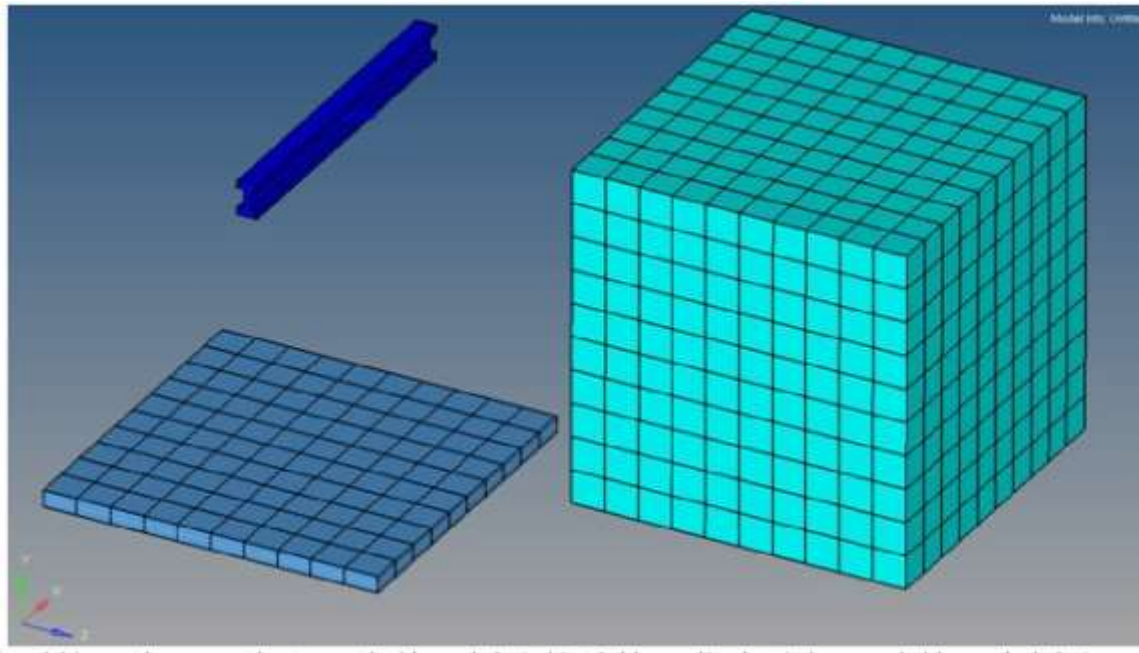
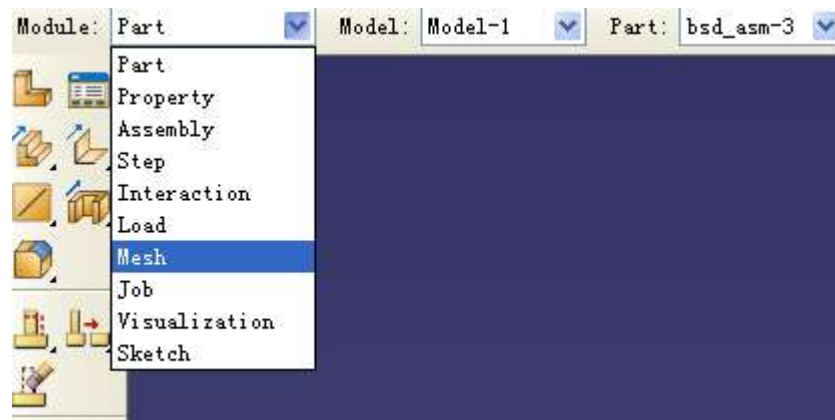
The contact stiffness is calculated by the following equation:

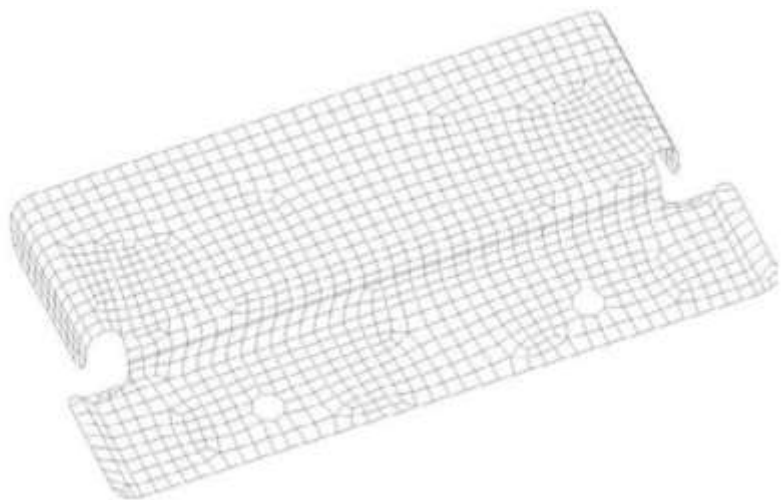
$$F = k \Delta$$

5 Load and Boundary conditions

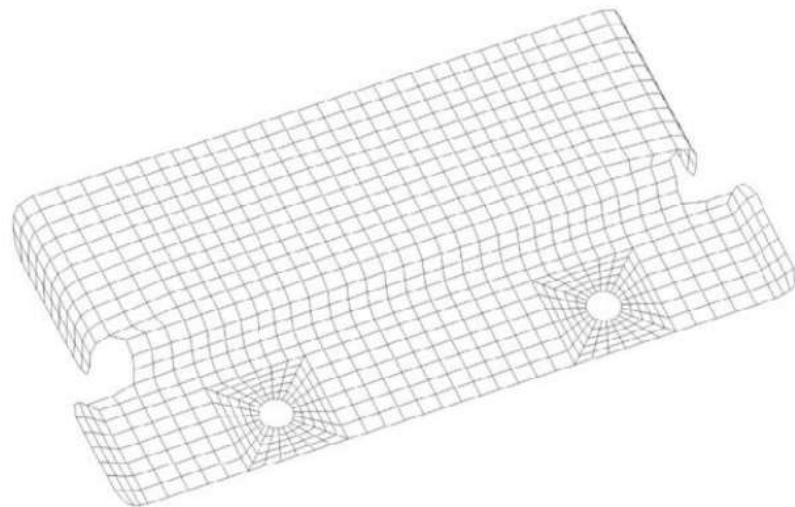


6 Mesh

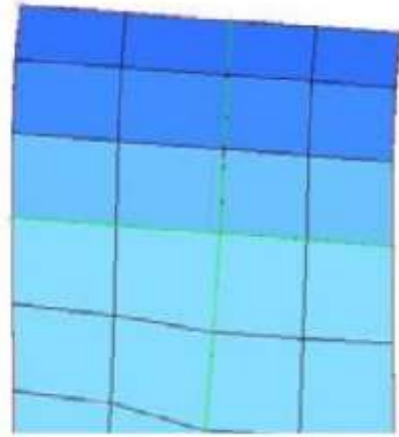
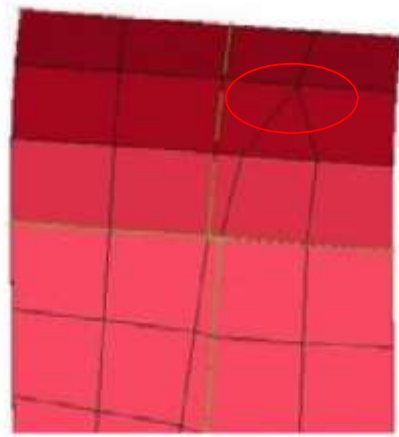
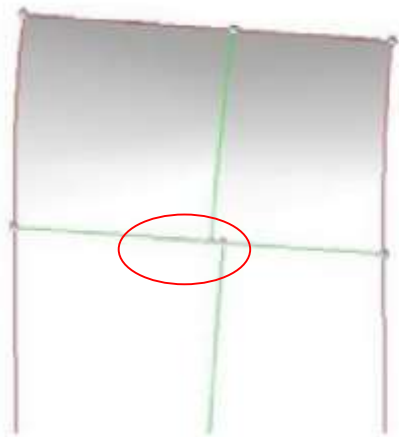




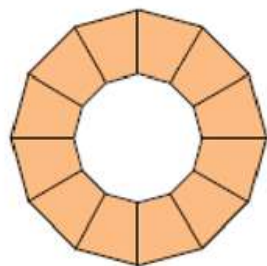
节点 = 1400, 单元 = 1309



节点 = 1073, 单元 = 982

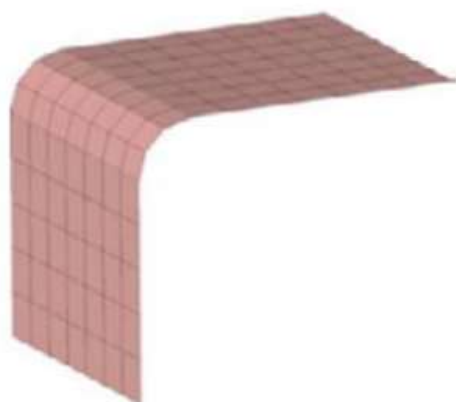
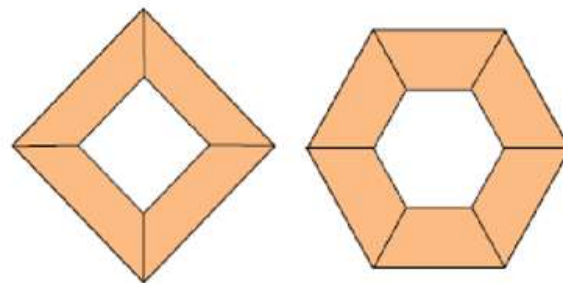


Refined mesh

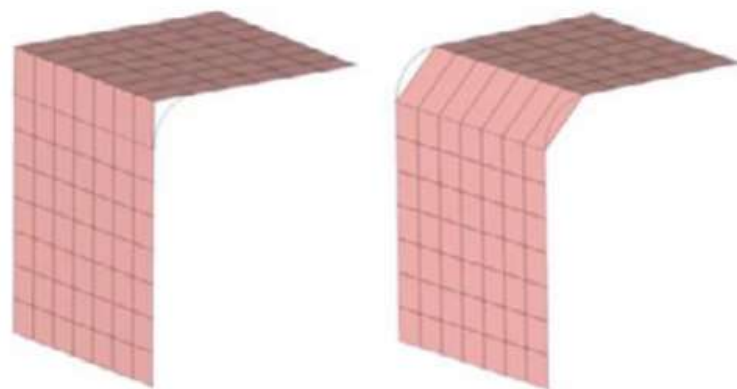


around bolt hole

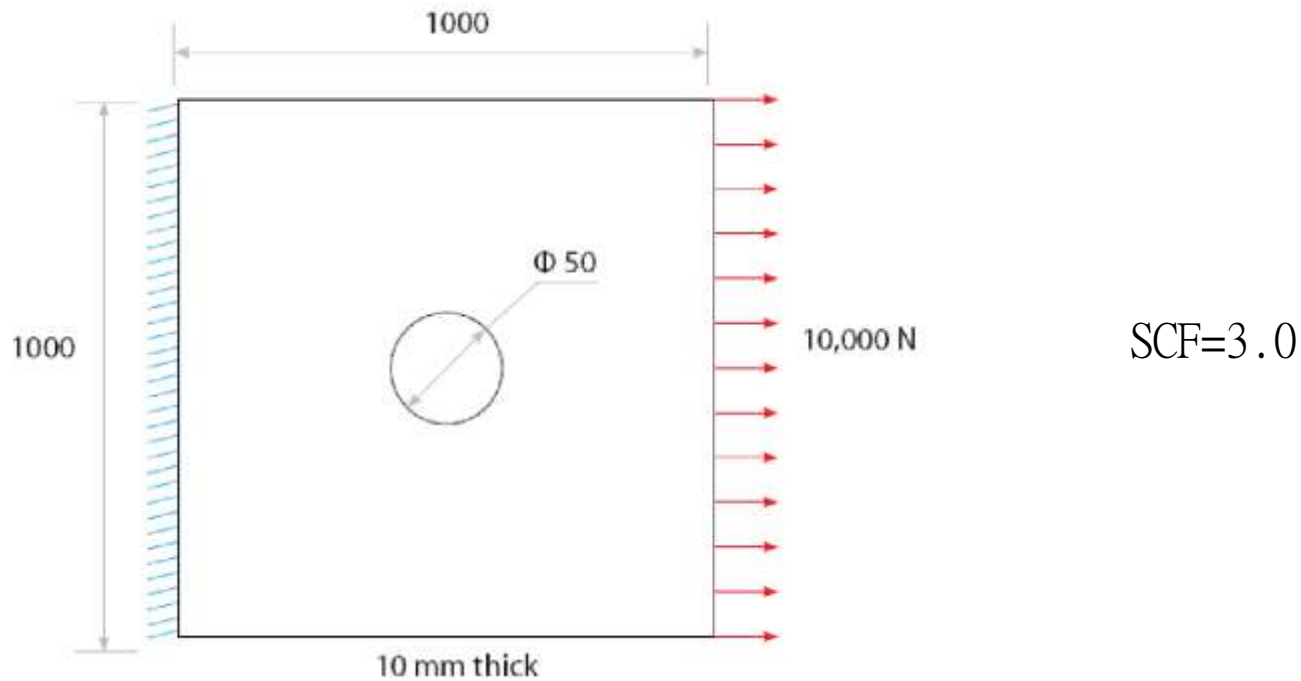
Normal mesh



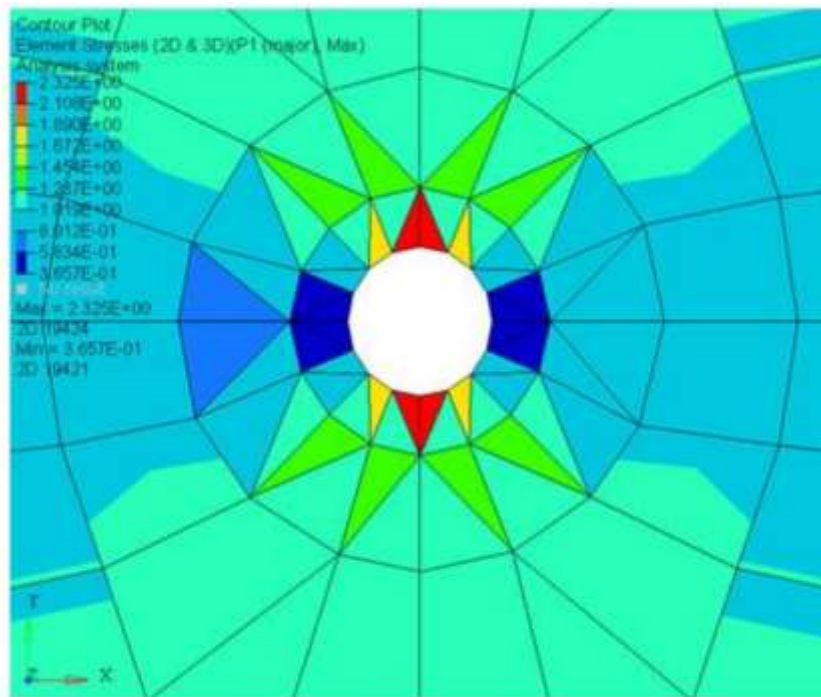
corner area



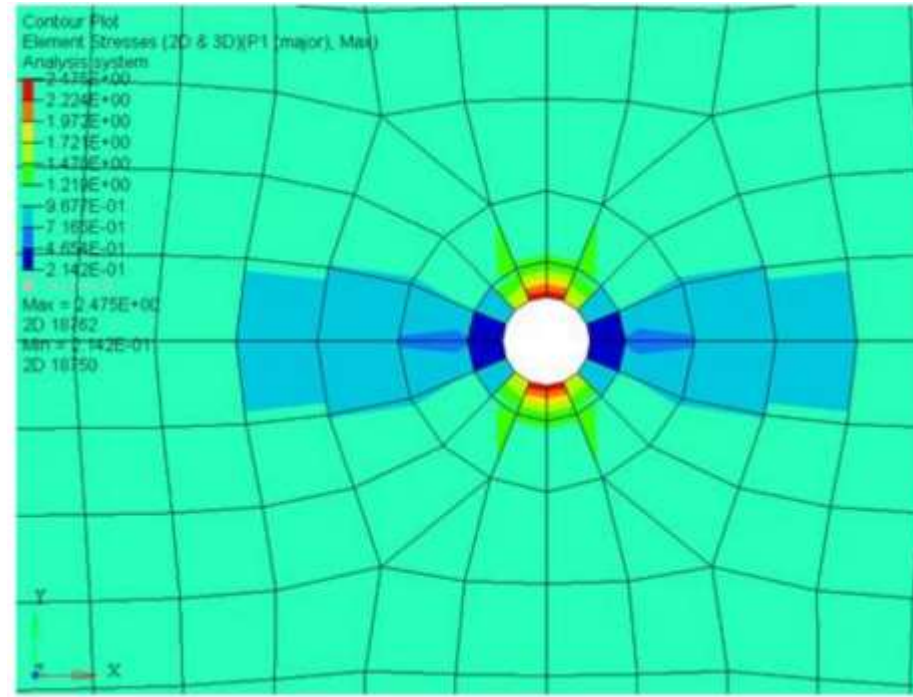
Case 1: a cantilever beam with the section of 250x20x5 mm subjected to a 35N load



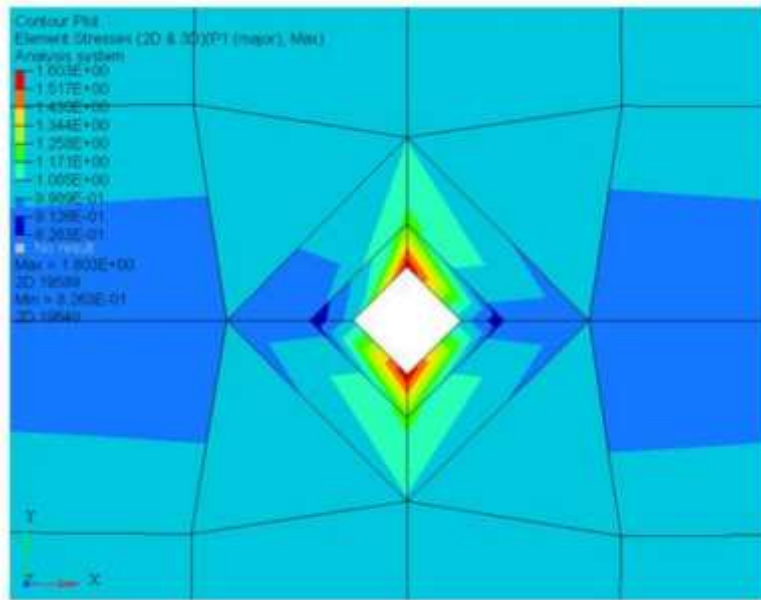
- ① Effect of element shape: triangle element and rectangular element;
- ② Influence of element number.



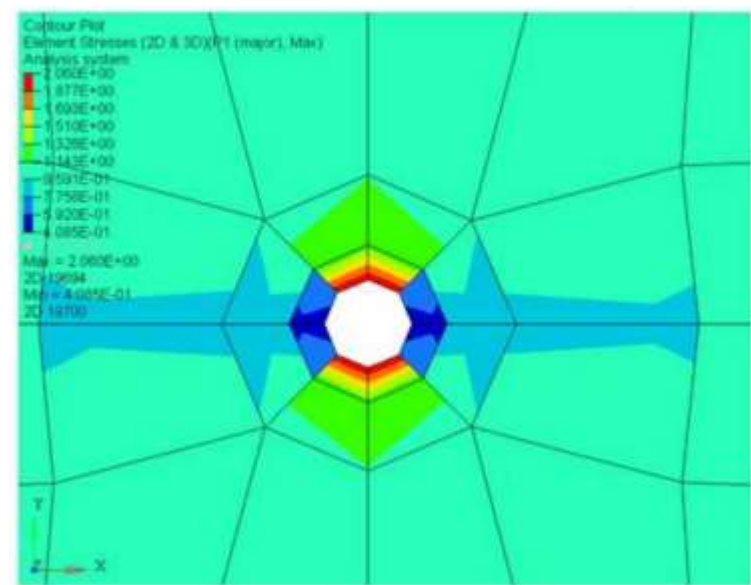
Maximum principle stress=2.32
MPa



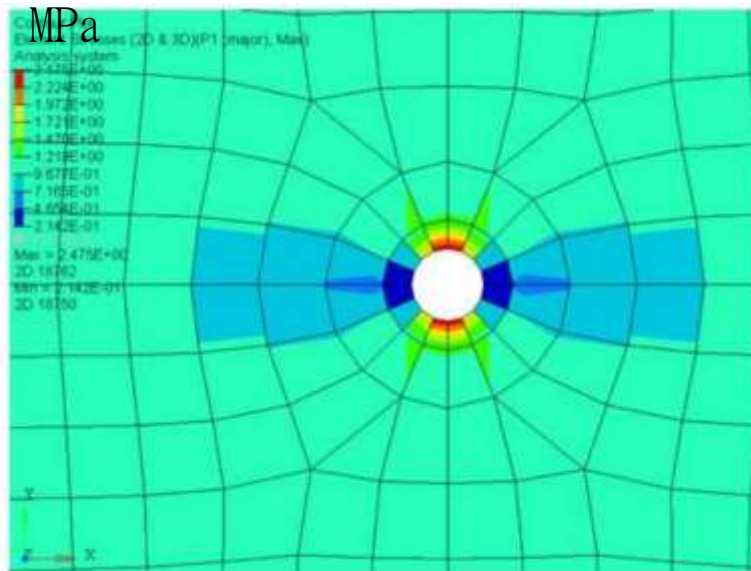
Maximum principle stress=2.47
MPa



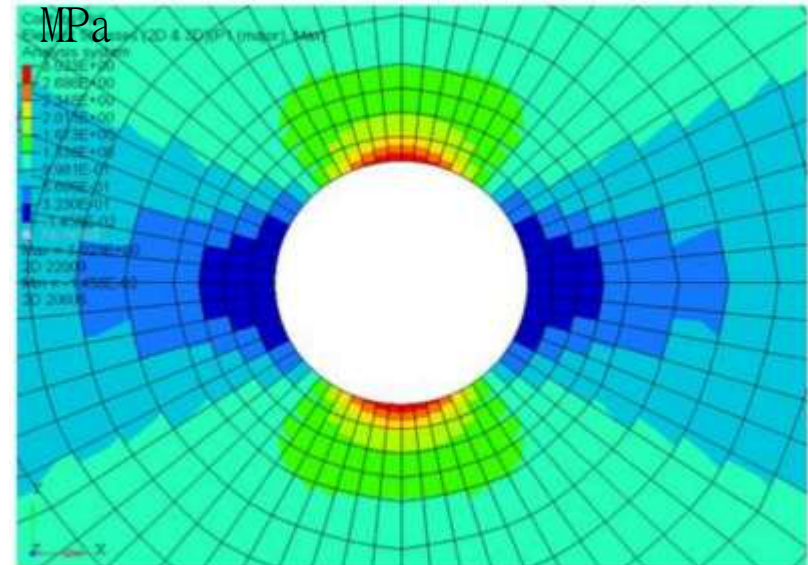
Element number on bolt hole=4
Maximum principle stress=1.6
MPa



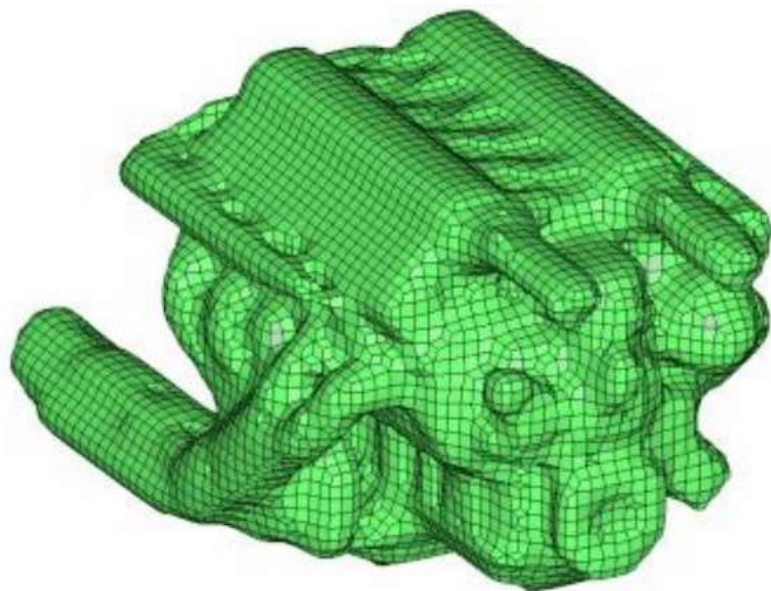
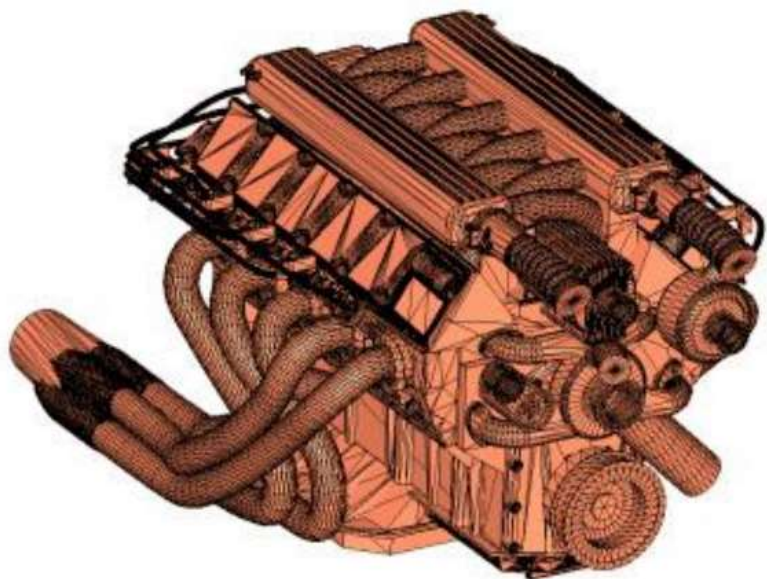
Element number on bolt hole=8
Maximum principle stress=2.0
MPa



Element number on bolt hole=16
Maximum principle stress=2.47
MPa



Element number on bolt hole=64
Maximum principle stress=3.02
MPa



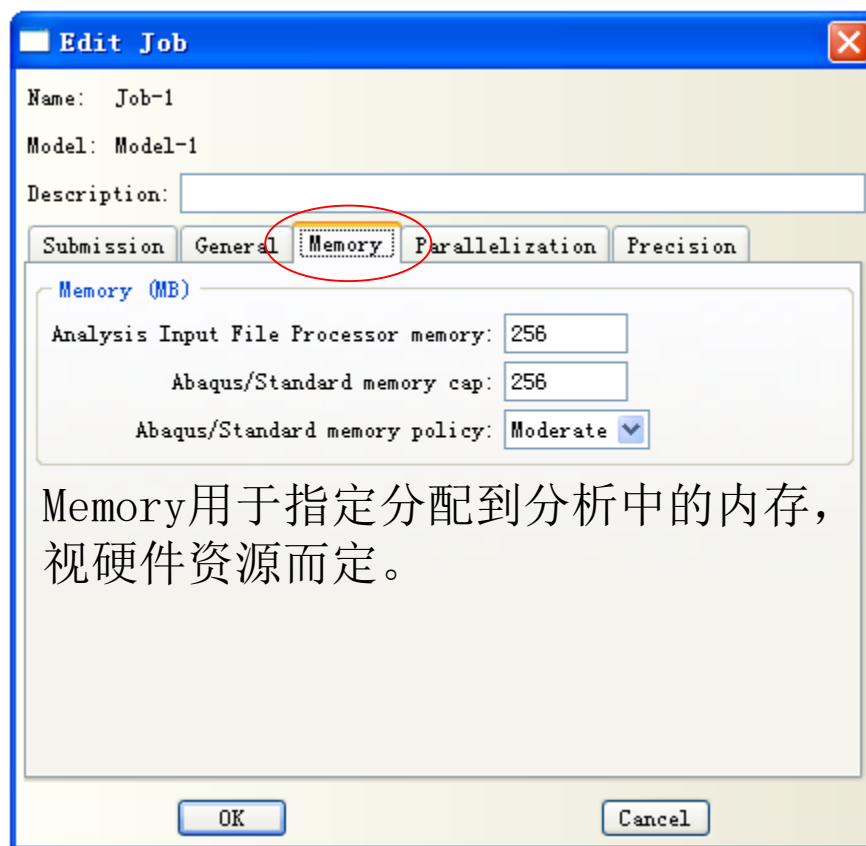
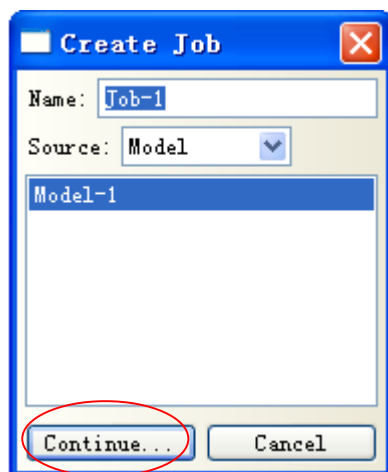
7 Step

(1) General analysis step:

- Static, General
- Dynamic Explicit
- Dynamic Implicit
- Heat transfer
- Coupled Temp-displacement
- Dynamic, Temp-disp, Explicit
- Mass diffusion
-

(2) Linear perturbation step (线性摄动分析) :

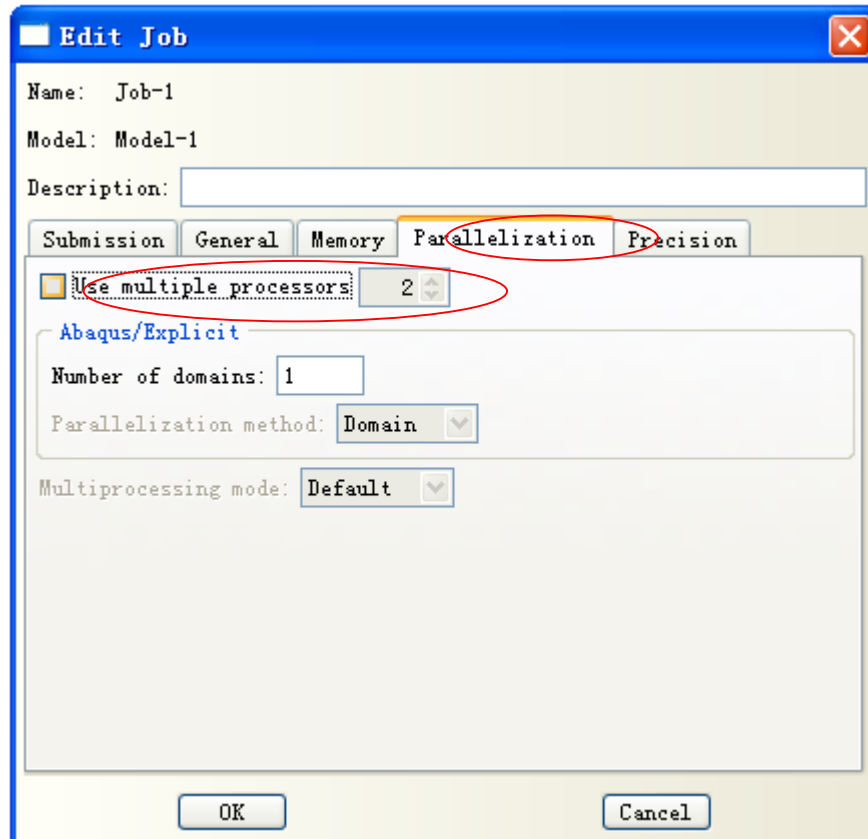
- Buckle
- Frequency
- Static Linear perturbation
- Steady-state dynamics, direct



Memory用于指定分配到分析中的内存，视硬件资源而定。

Abaqus/Standard memory policy当分配的内存大于实际分析所需的内存，多余内存的使用设置。

- Minimum: 闲置
- Moderate: 该选项通常能自动提供合理的内存使用，建议采用此默认设置
- Maximum: 将多余内存都用于存储临时文件



Parallelization用于并行运算的设置

Use multiple processors设置用于分析的处理器数目，默认值为1，即不使用并行运算。

9 Visualization



} Display options

} Output results

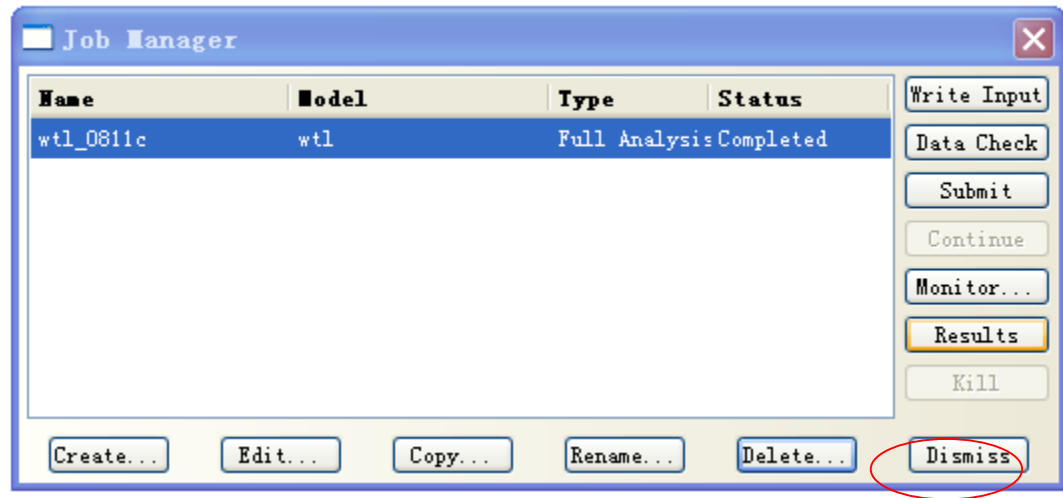
} Animation

} Coordinates

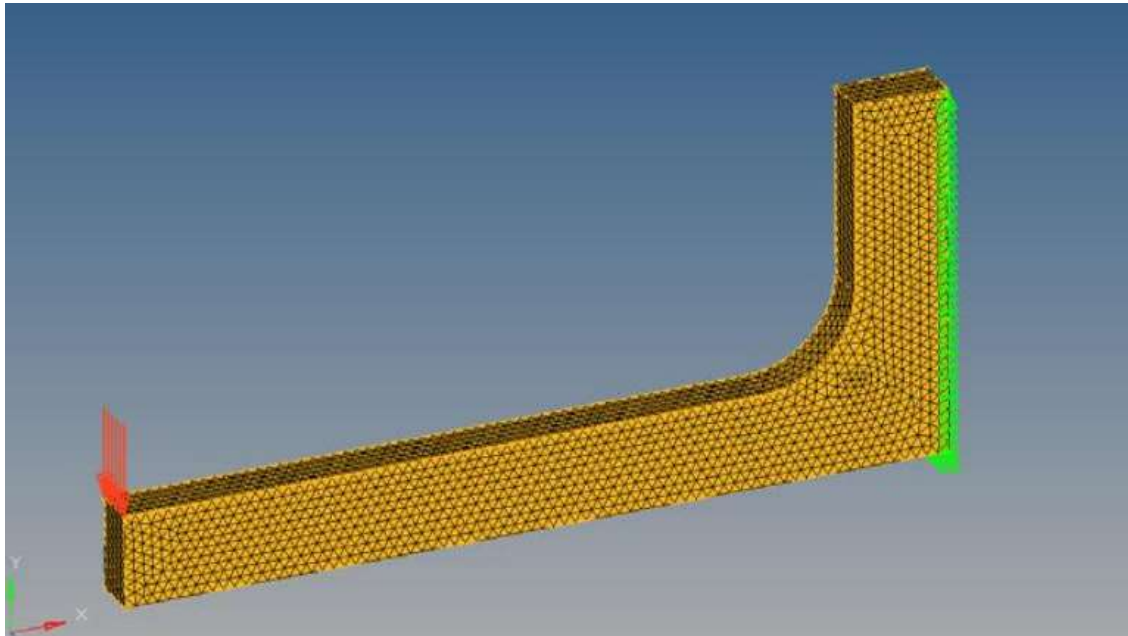
} Curves and tables

$$\}$$

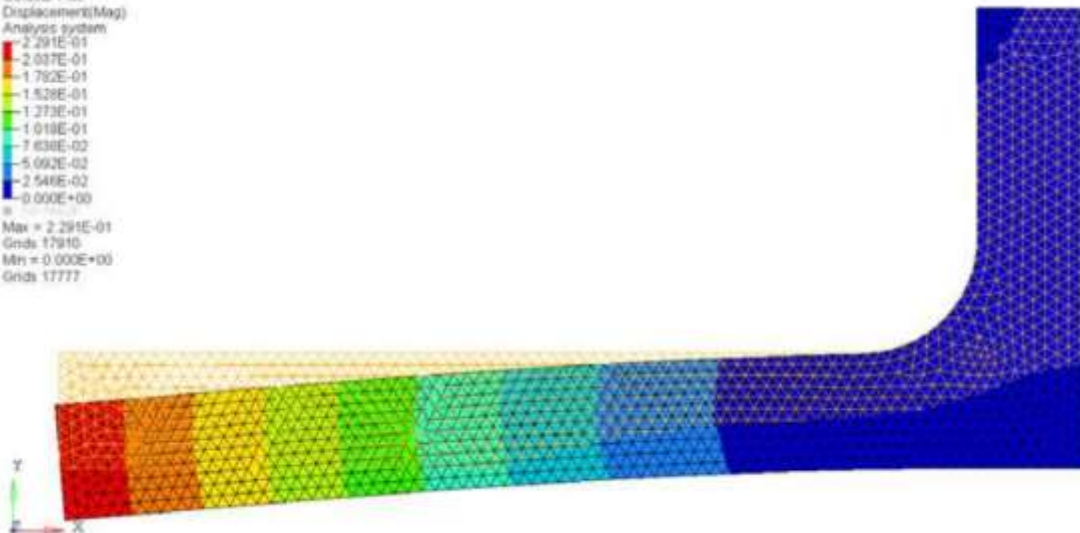
} View cut

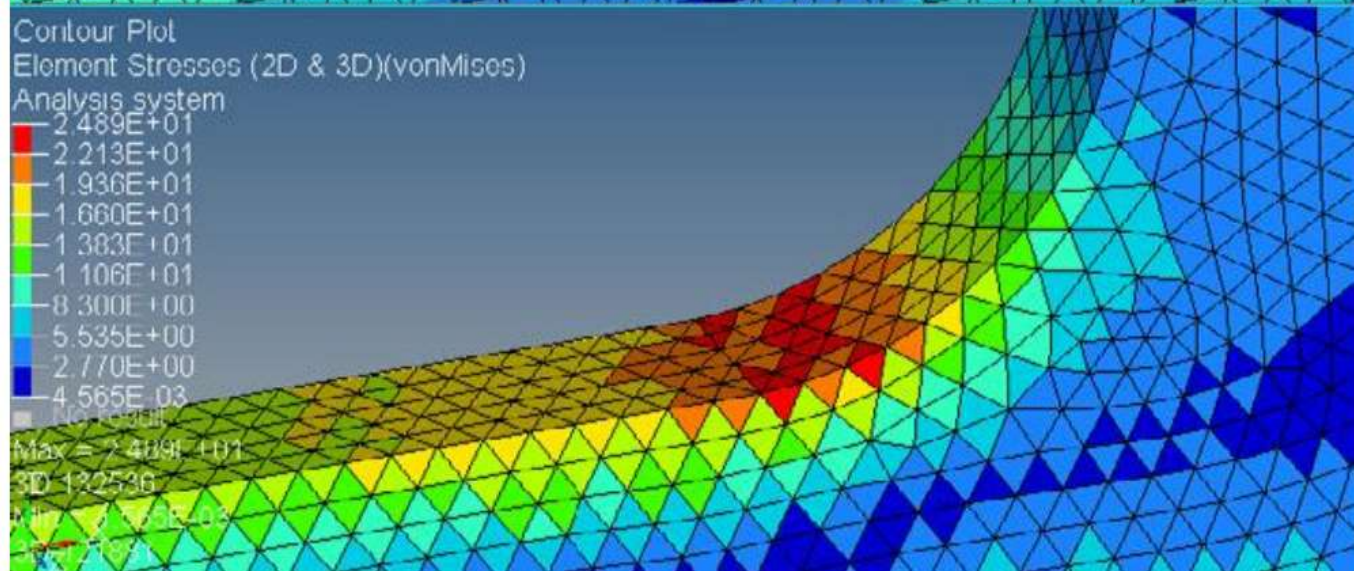
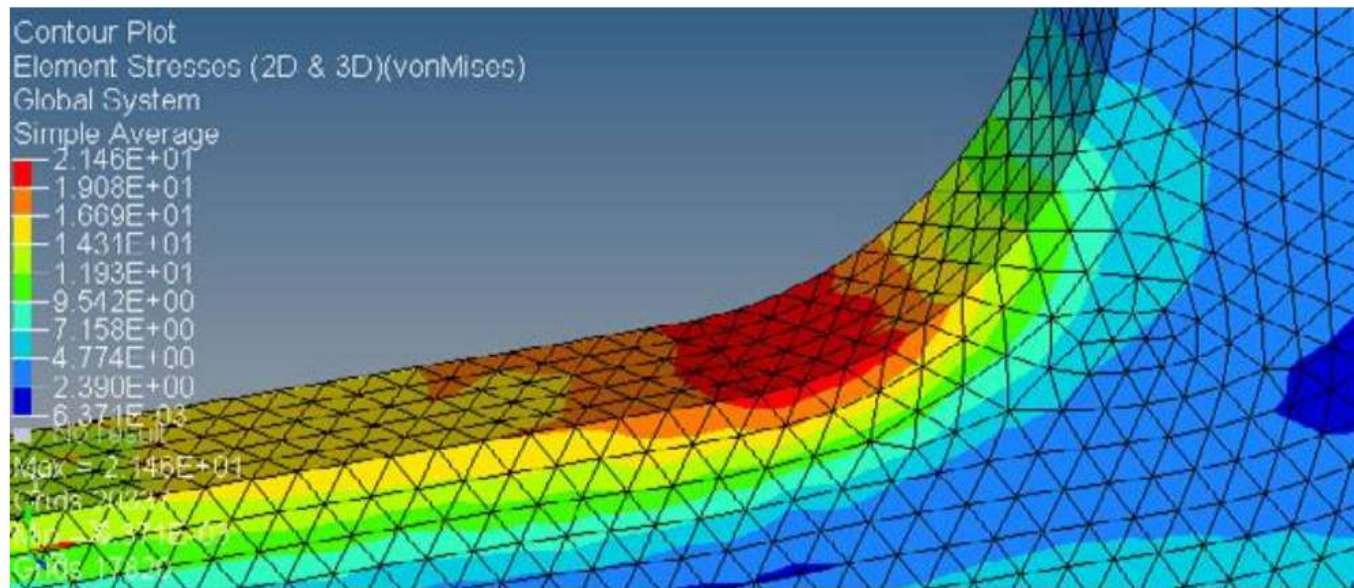


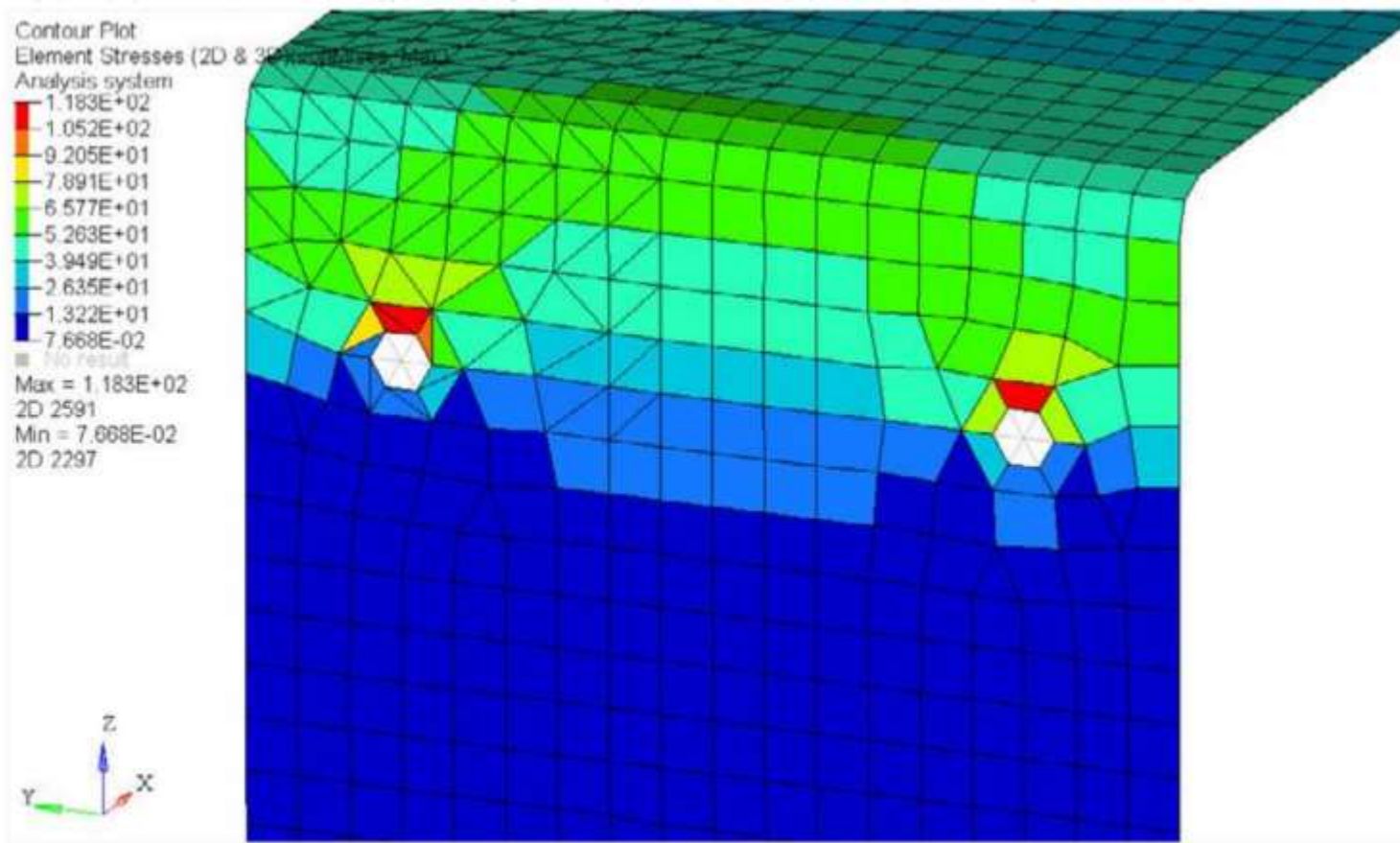
Contour output



Contour Plot
Displacement(Mag)
Analysis system
2.291E-01
2.037E-01
1.782E-01
1.528E-01
1.273E-01
1.018E-01
7.638E-02
5.092E-02
2.546E-02
0.000E+00
Max = 2.291E-01
Gnds 17910
Min = 0.000E+00
Gnds 17777







References

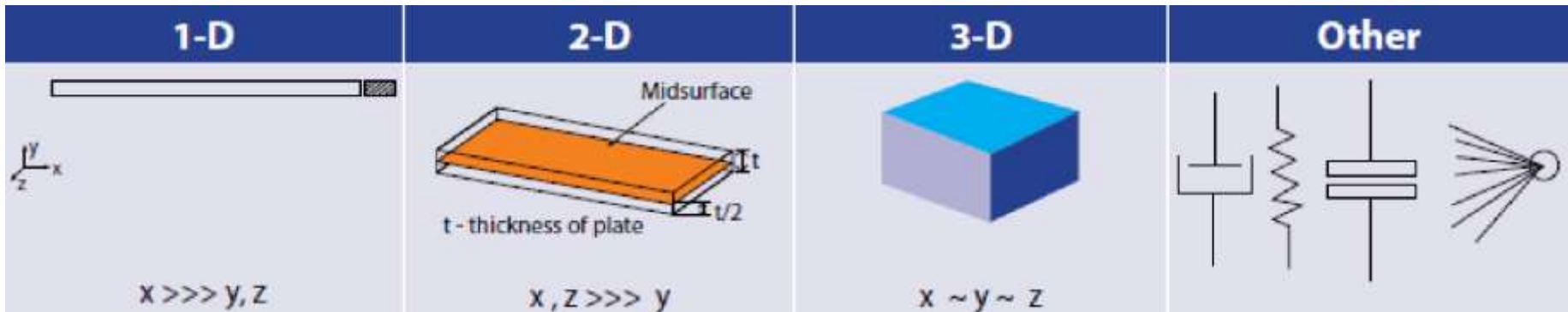
- ABAQUS Users Manual
- ABAQUS非线性有限元分析实例
- Examples

ANSYS

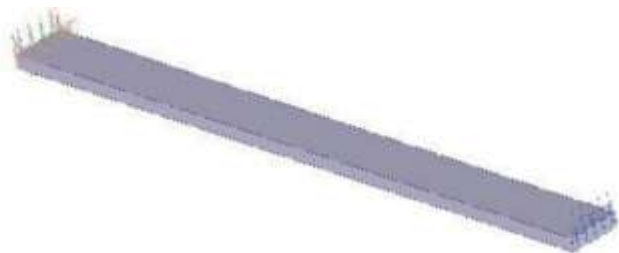
- 1 Element database
- 2 Preprocessor and postprocessor
- 3 APDL modelling method
- 4 Nonlinear analysis
- 5 ANSYS Workbench

ANSYS element database

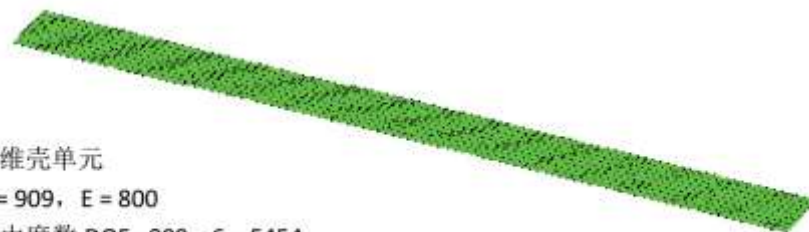
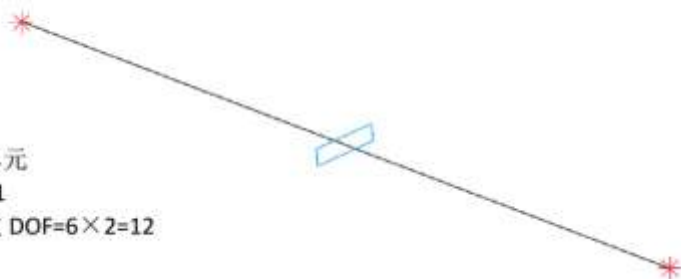
- BEAM ELEMENT 梁/管单元
- SHELL ELEMENT 壳/膜单元
- SOLID ELEMENT 实体单元
- LINK ELEMENT 杆/索单元
- SPRING ELEMENT 弹簧元
- CONTACT ELEMENT 接触单元
- MASS ELEMENT 质量单元
-



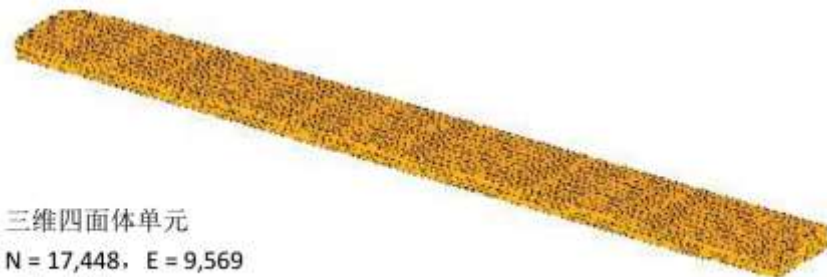
Case 2: a cantilever beam with the section of 250x20x5 mm subjected to a 35N load.



一维梁单元
 $N=2$, $E=1$
 自由度数 $DOF=6 \times 2=12$



二维壳单元
 $N = 909$, $E = 800$
 自由度数 $DOF= 909 \times 6 = 5454$



三维四面体单元
 $N = 17,448$, $E = 9,569$
 自由度数 $DOF= 17,448 \times 3 = 52,344$

	节点数量	单元数量	应力(N/mm ²)	位移(mm)
理论解	--	--	105	4.23
一维单元	2	1	105	4.23
二维单元	909	800	103	4.21
三维单元	17,448	9,569	104	4.21

Beam

Linear analysis 线性分析

- *It recommends to use Beam3, Beam4, Beam188 and Beam189 in linear analysis.*

Nonlinear analysis 非线性分析

- *Beam188 and Beam189 can be applied to simulate isotropic plastic, large strain, buckling and large rotation problems.*



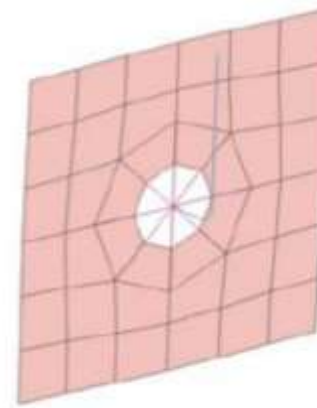
Shell

Linear analysis 线性分析

- *Shell163 is a suit for the thin-walled sections, it has flexural and membrane effects but cannot resist the shear force.*

- *Shell143, 143, 181, 91, 93 and 99 are thick wall elements*

Nonlinear analysis 非线性分析
which can consider the shear strain. Shell181 is a commonly used nonlinear shell element, it can fulfill a extensive analysis requirements with a limit CPU time



Material properties

- *Isotropic:*

E_X

ν_{XY}

$$G_{XY} = E_X / (2(1 + \nu_{XY}))$$



- *Anisotropic:*

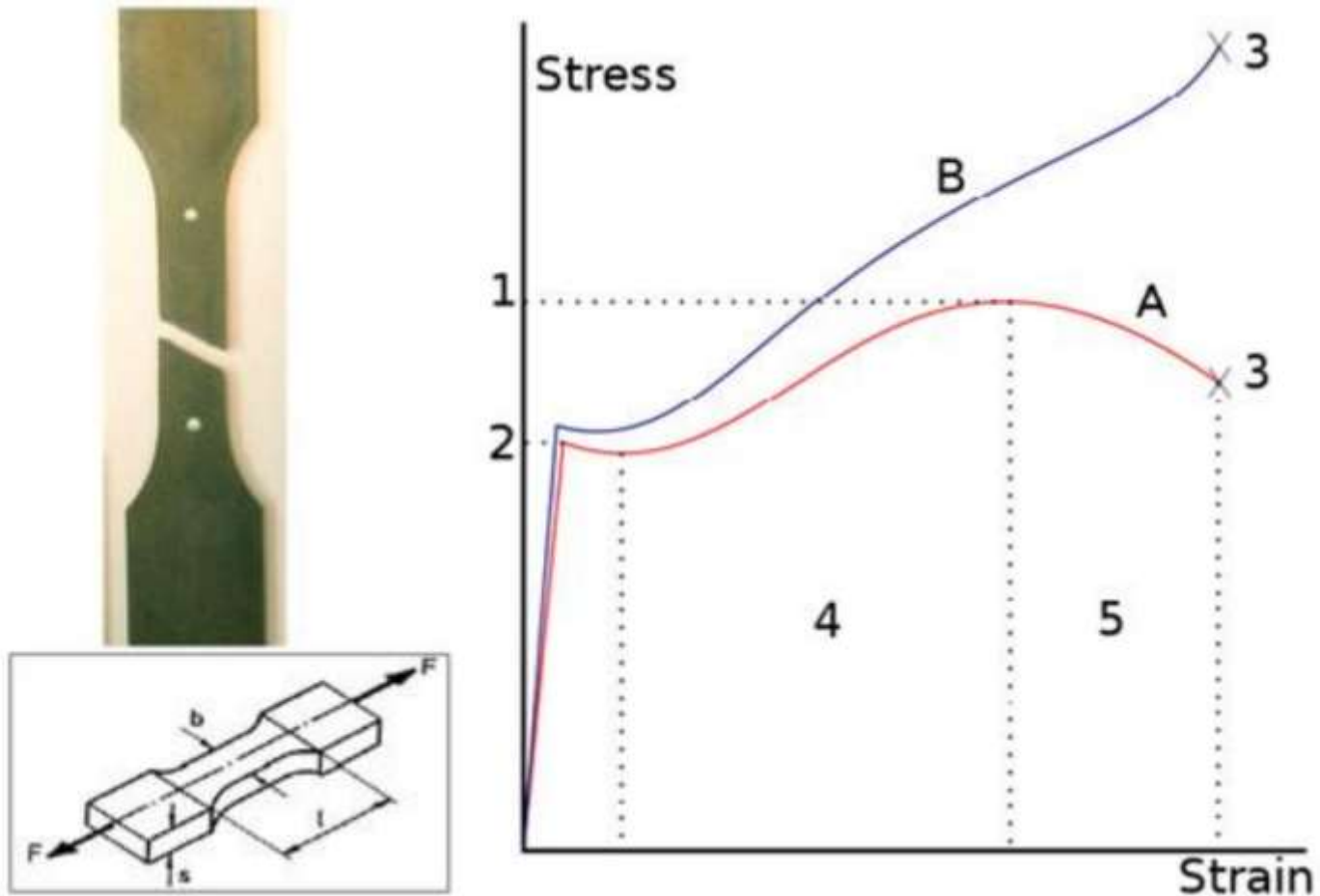
E_X, E_Y, E_Z

$\nu_{XY}, \nu_{YZ}, \nu_{XZ}$

$G_{XY}, G_{YZ}, \text{ and } G_{XZ}$



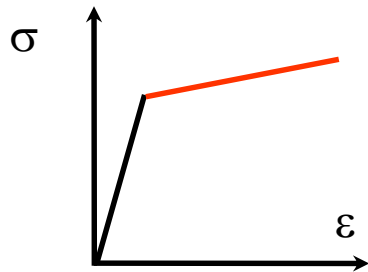
Stress-strain curve



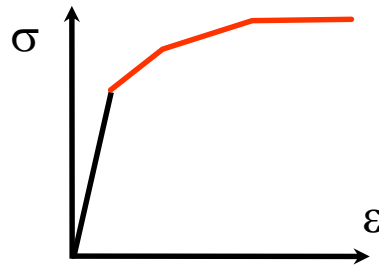
$$\sigma_{tr} = \sigma_e \exp(\epsilon_{tr}) \quad \epsilon_{tr} = \ln(1 + \epsilon_e)$$

$$\sigma_{tr} = \sigma_e (1 + \epsilon_e)$$

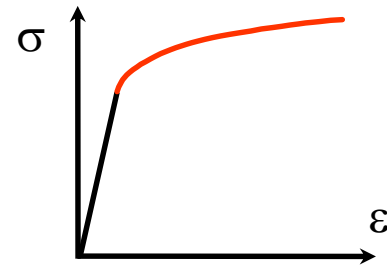
- ANSYS plastic material including three stress-strain curves:



Bilinear



Multi-linear



Nonlinear

Yield criterion

To choose the proper criterion for determine the yielding of the element

Von Mises criterion

- Von Mises equivalent stress follows the following equation:

$$\sigma_e = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]}$$

where σ_1 、 σ_2 and σ_3 are principal stress, the yield occurs when the equivalent stress beyond the yield stress :

$$\sigma_e > \sigma_y$$

Hill criterion

- Hill criterion used in anisotropic models, it follows the following equation:

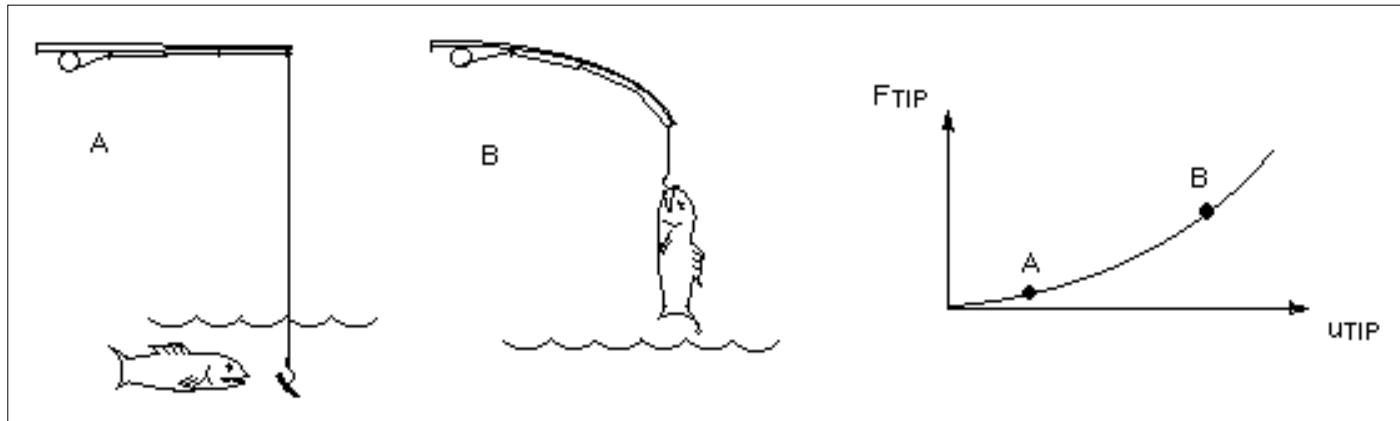
$$\sigma_o = \sqrt{H(\sigma_x - \sigma_y)^2 + F(\sigma_y - \sigma_z)^2 + G(\sigma_x - \sigma_z)^2 + 2N\tau_{xy}^2 + 2L\tau_{yz}^2 + 2M\tau_{xz}^2}$$

where R_{xx} , R_{yy} , R_{zz} , R_{xy} , R_{yz} , R_{xz} can be achieved as following formulas:

$$\begin{array}{llll} G + H = \frac{1}{R_{xx}^2} & N = \frac{3}{2R_{xy}^2} & R_{xx} = \frac{\sigma_{xx}^y}{\sigma_o} & R_{xy} = \sqrt{3} \frac{\sigma_{xy}^y}{\sigma_o} \\ F + H = \frac{1}{R_{yy}^2} & L = \frac{3}{2R_{yz}^2} & R_{yy} = \frac{\sigma_{yy}^y}{\sigma_o} & R_{yz} = \sqrt{3} \frac{\sigma_{yz}^y}{\sigma_o} \\ F + G = \frac{1}{R_{zz}^2} & M = \frac{3}{2R_{xz}^2} & R_{zz} = \frac{\sigma_{zz}^y}{\sigma_o} & R_{xz} = \sqrt{3} \frac{\sigma_{xz}^y}{\sigma_o} \end{array}$$

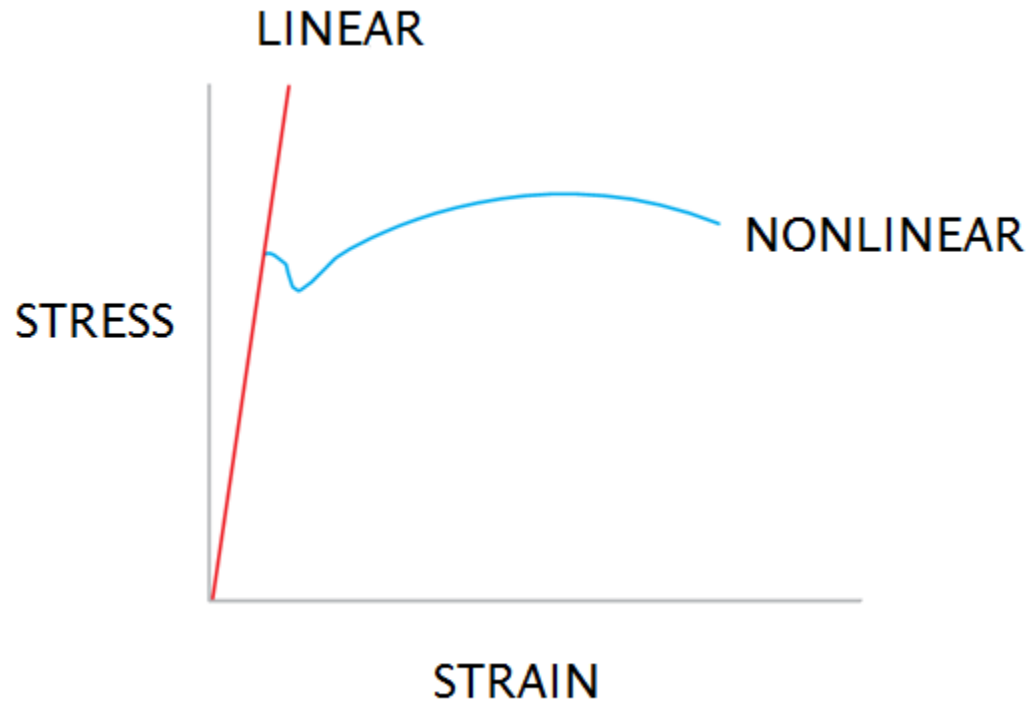
- Geometrical nonlinear

- If a structure has experienced a large deformation, the change of geometry can cause nonlinear behavior.



- **Material nonlinear**

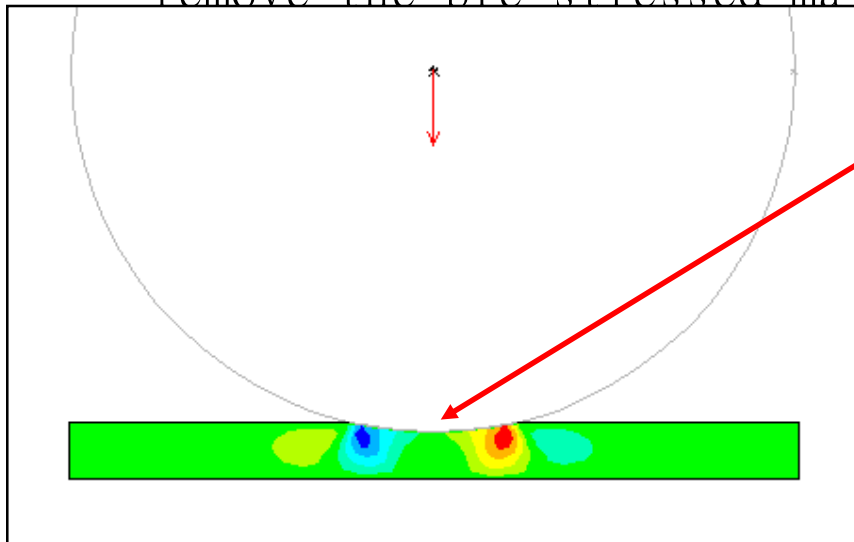
- The nonlinear stress-strain relationship is the common cause of nonlinear structural behaviour.



- State nonlinear

- State nonlinear is another cause of nonlinear behaviour, for example:

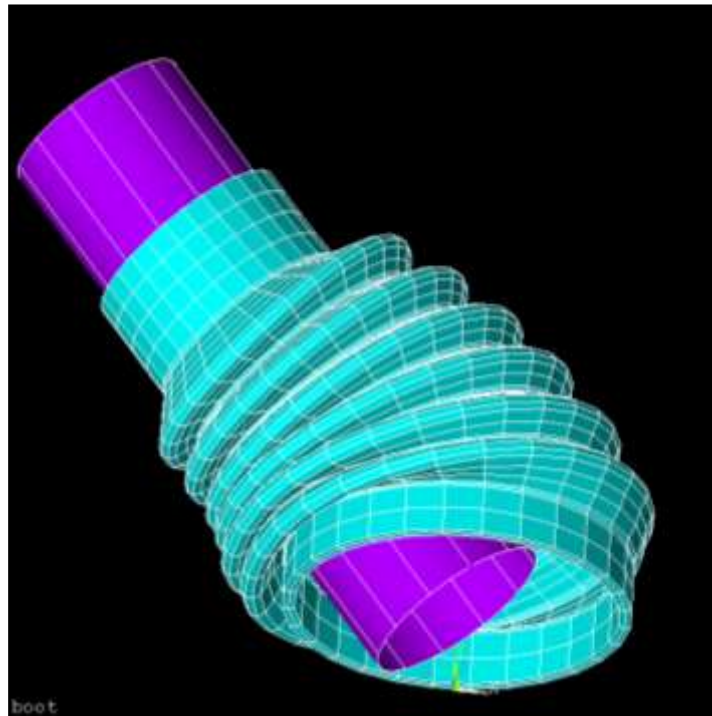
- cable changes from the relaxation to the tensioning state
 - two parts in an assembly suddenly come into the contact state
 - remove the pre-stressed materials during manufacturing



In this case, the contact state changes from separate to close during the increasing loading.

- **Combined nonlinear**

Rubber dynamic analysis including geometrical, material, and state nonlinear.



APDL

- APDL is the abbreviation of **ANSYS parametric design language**, it is a kind of scripting language which can be used to parametrically complete the finite element analysis, is a principal tool for optimal design and adaptive meshing.
- APDL allows complex data input, user has control on every design or analysis, geometric dimensions material properties such as model

APDL has many advantages over traditional modelling method:

- ❖ it reduces modelling time;
- ❖ It can be saved in a quite limited size;
- ❖ It is relevant with the version of ANSYS;
- ❖ It is good for some advanced functions such as optimum analysis and adaptive;
- ❖ It can be used to establish a part database;
- ❖ It can be extended in the second development.

名称	功 能	路 径	命 令
prep7	建立几何模型，赋予材料属性，分网与施加边界条件等	Main Menu>Preprocessor	/prep7
solution	加载、求解	Main Menu>Solution	/solu
post1	查看某个时刻的计算结果	Main Menu>General Postproc	/post1
post26	查看时间历程上的计算结果	Main Menu>TimeHist Postpro	/post26
opt	优化设计	Main Menu>Design Opt	/opt
pds	概率设计	Main Menu>Prob Design	/pds
aux2	把二进制文件变为可读文件	Utility Menu>File>List>Ninary Files	/aux2
aux12	在热分析中计算辐射因子和矩阵	Main Menu>Radiation Opt	/aux12
aux15	从CAD或FEM程序中传递文件	Main Menu>File>Import	/aux15
runstat	估计计算时间、运行状态等	Main Menu>Run-Time Stats	/runst

APDL

...language

- example:

jobname='proj1'

/filnam,jobname ! 作业名

/prep7

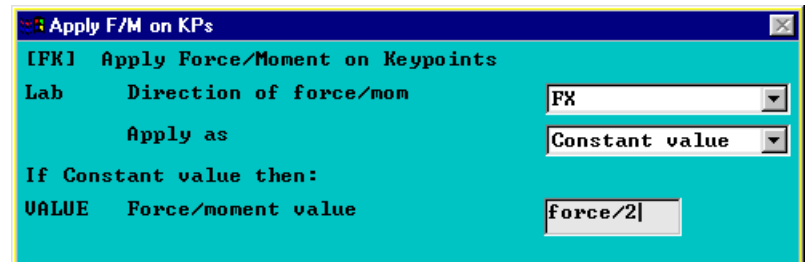
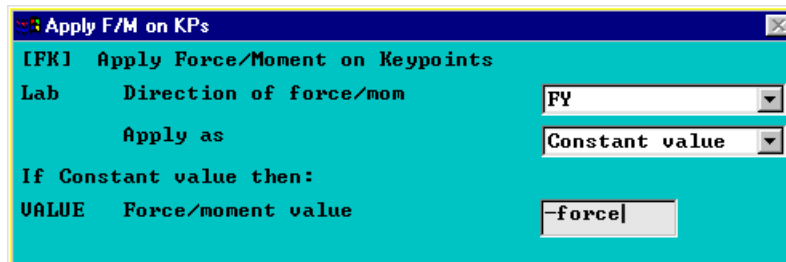
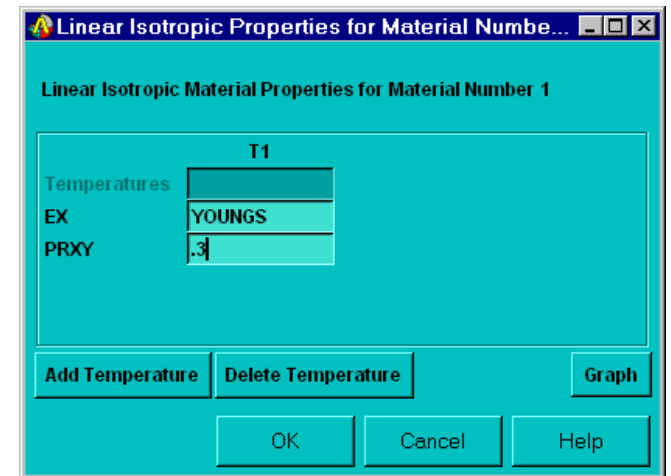
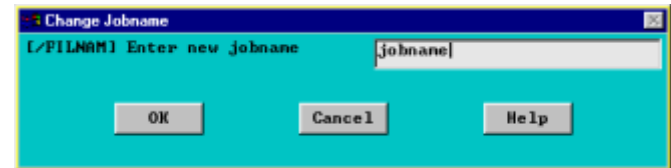
youngs=30e6

mp,ex,1,youngs ! 杨氏模量

force=500

fk,2,fy,-force ! 2号关键点的力

fk,6,fx,force/2 ! 6号关键点的力



References

- ANSYS工程结构数值分析 - 王新敏
- ANSYS土木工程应用实例
- ANSYS Users Manual
- Examples

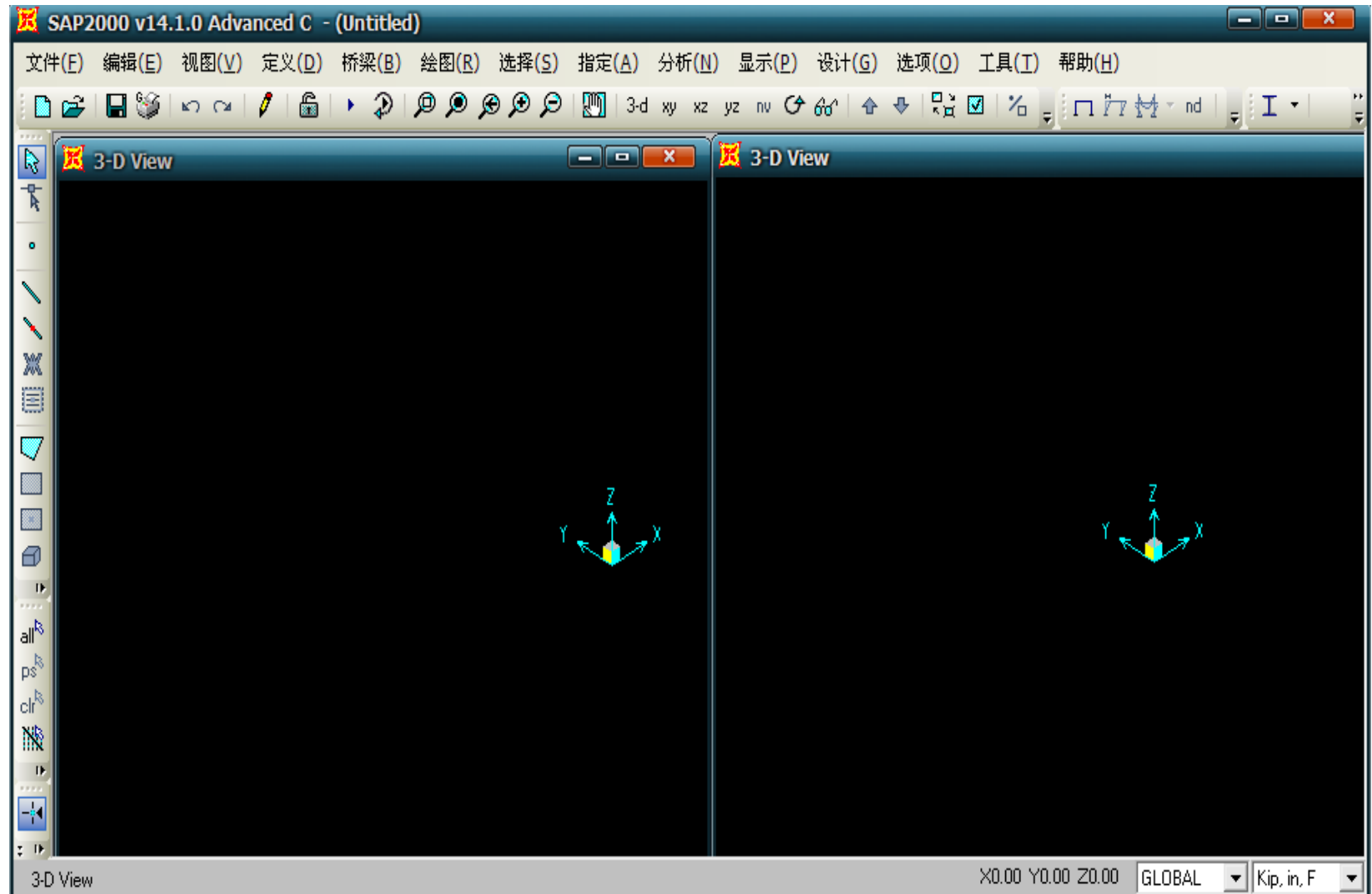
SAP2000
(FE Design program)

Features

- ❑ Space structures 复杂的空间结构检算
- ❑ Concrete, steel, aluminum frames 混凝土框架、钢结构框架、铝框架
- ❑ Industrial facilities 工业结构设计

- Design software 面向对象的设计软件
- 3D modelling 三维功能
- Bridge module 桥梁模块
- Advanced technique (large displacement, temperature, buckling and blast, etc.) 先进的分析技术（大变形、温度、拉压、屈服、爆炸等）

Interface

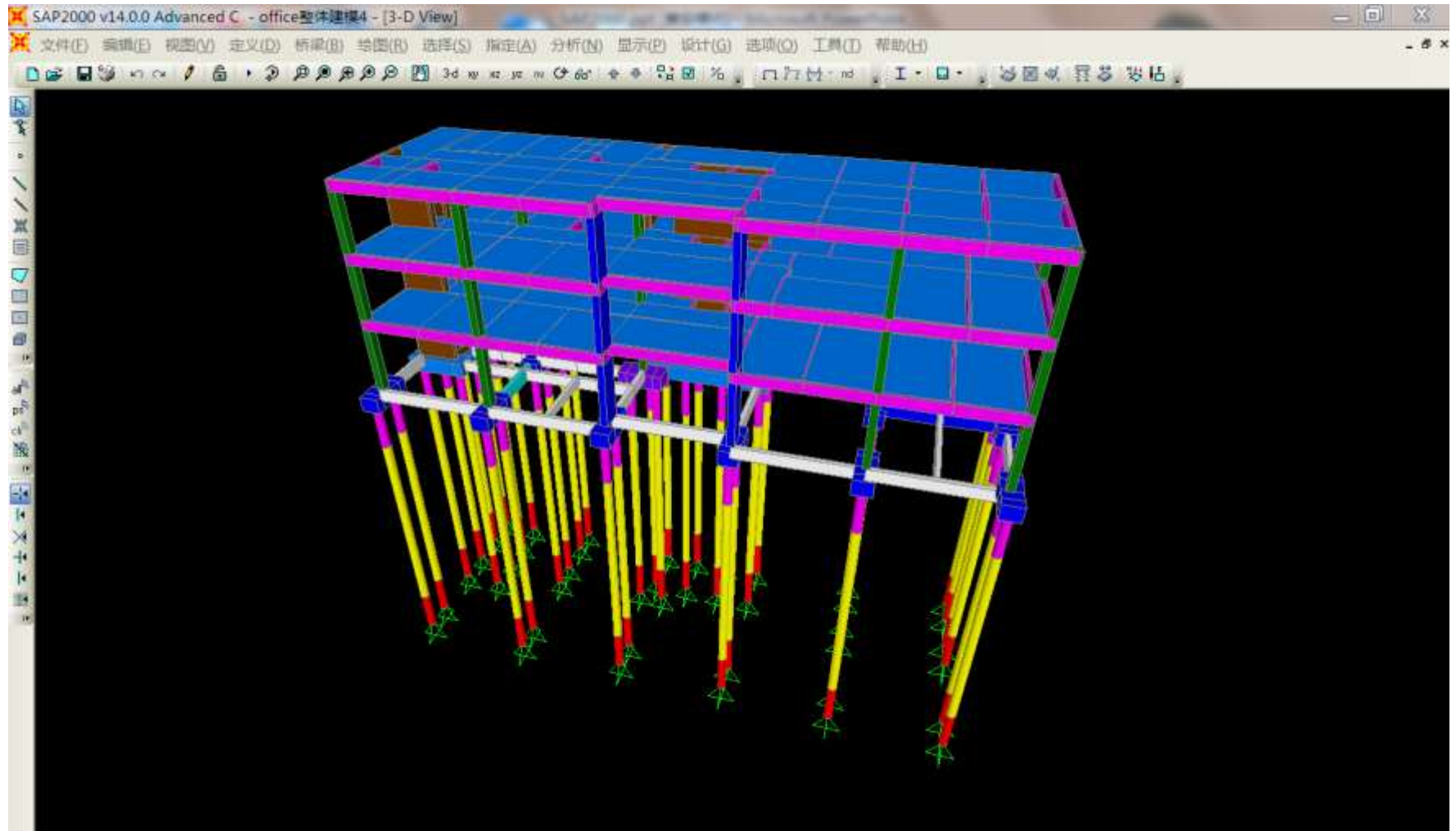


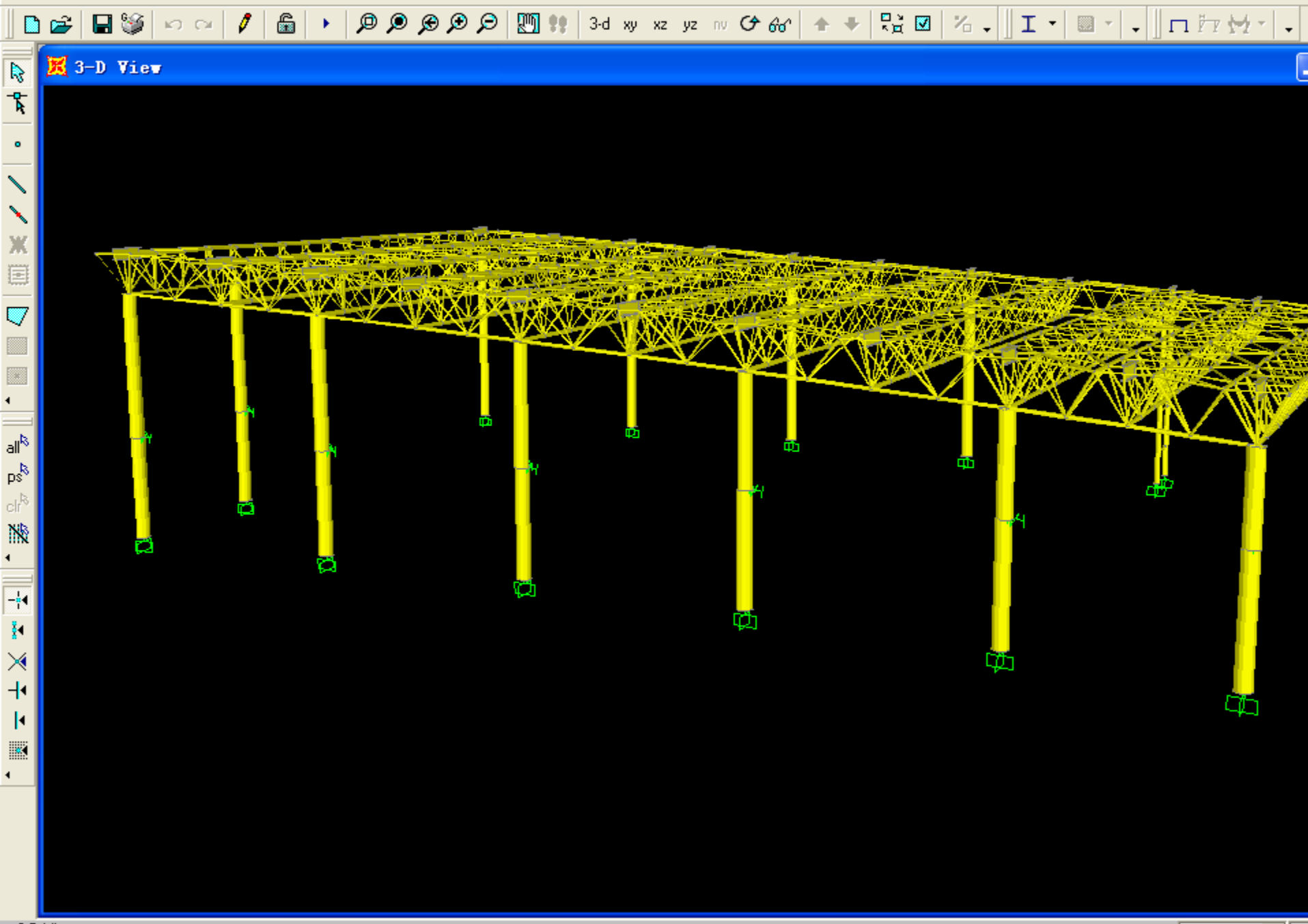
Modelling environment

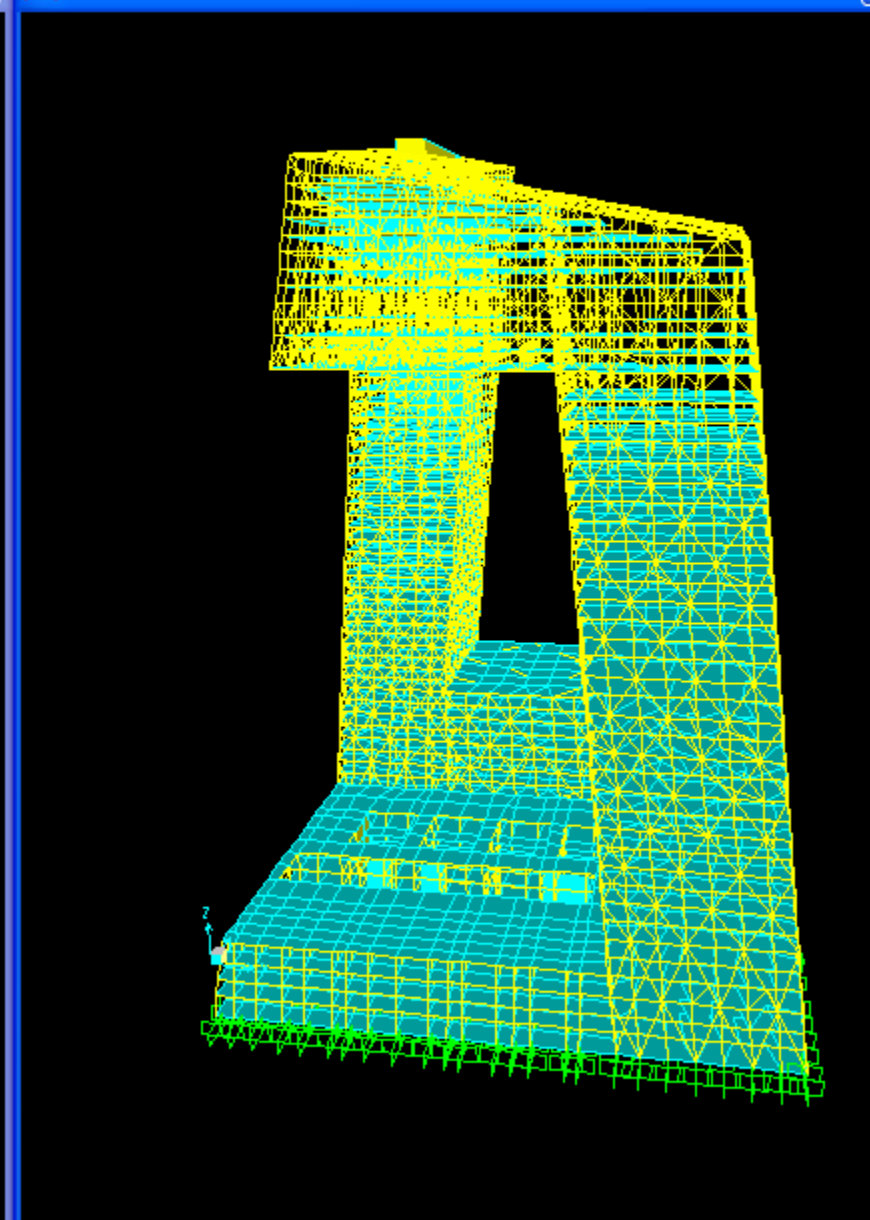
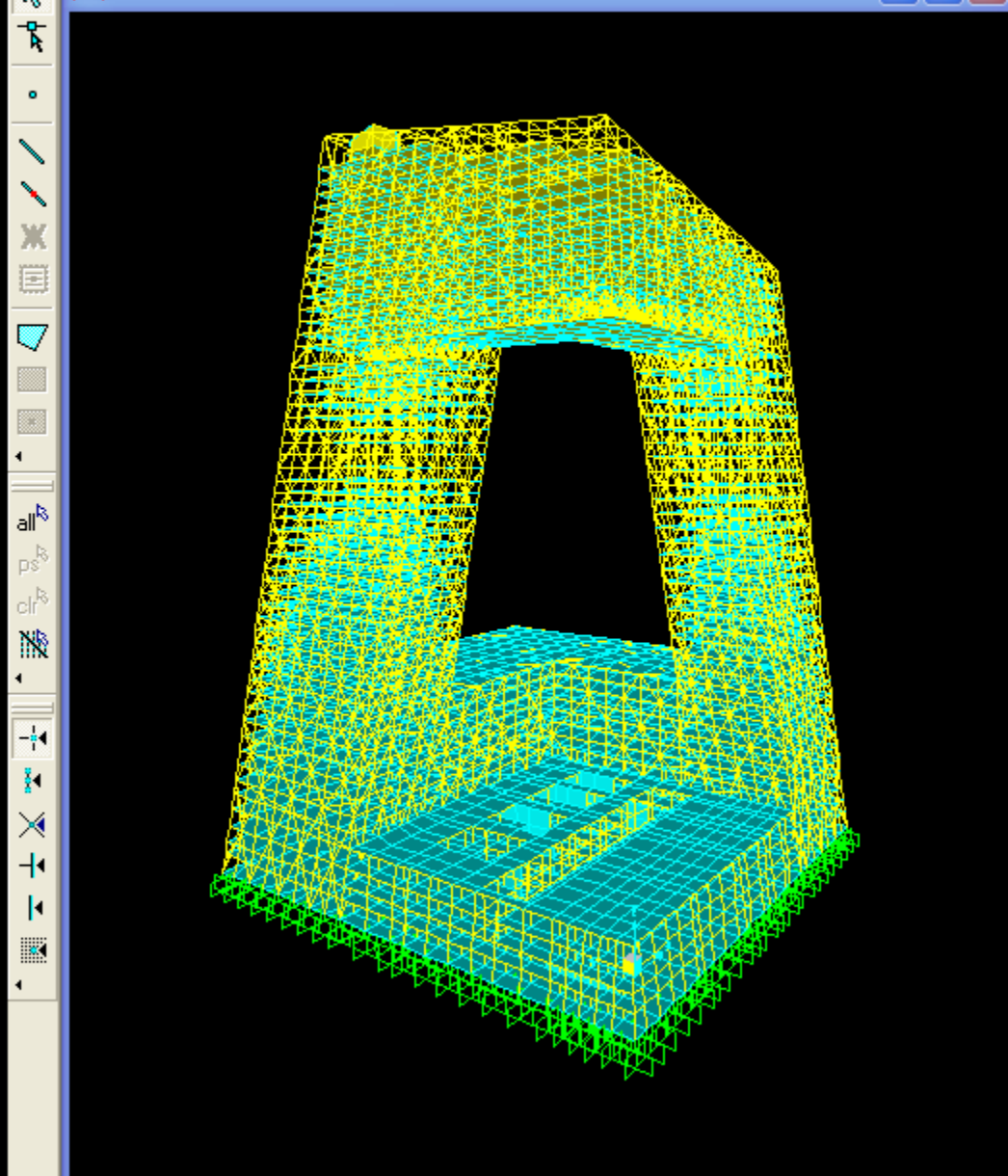
1 templates

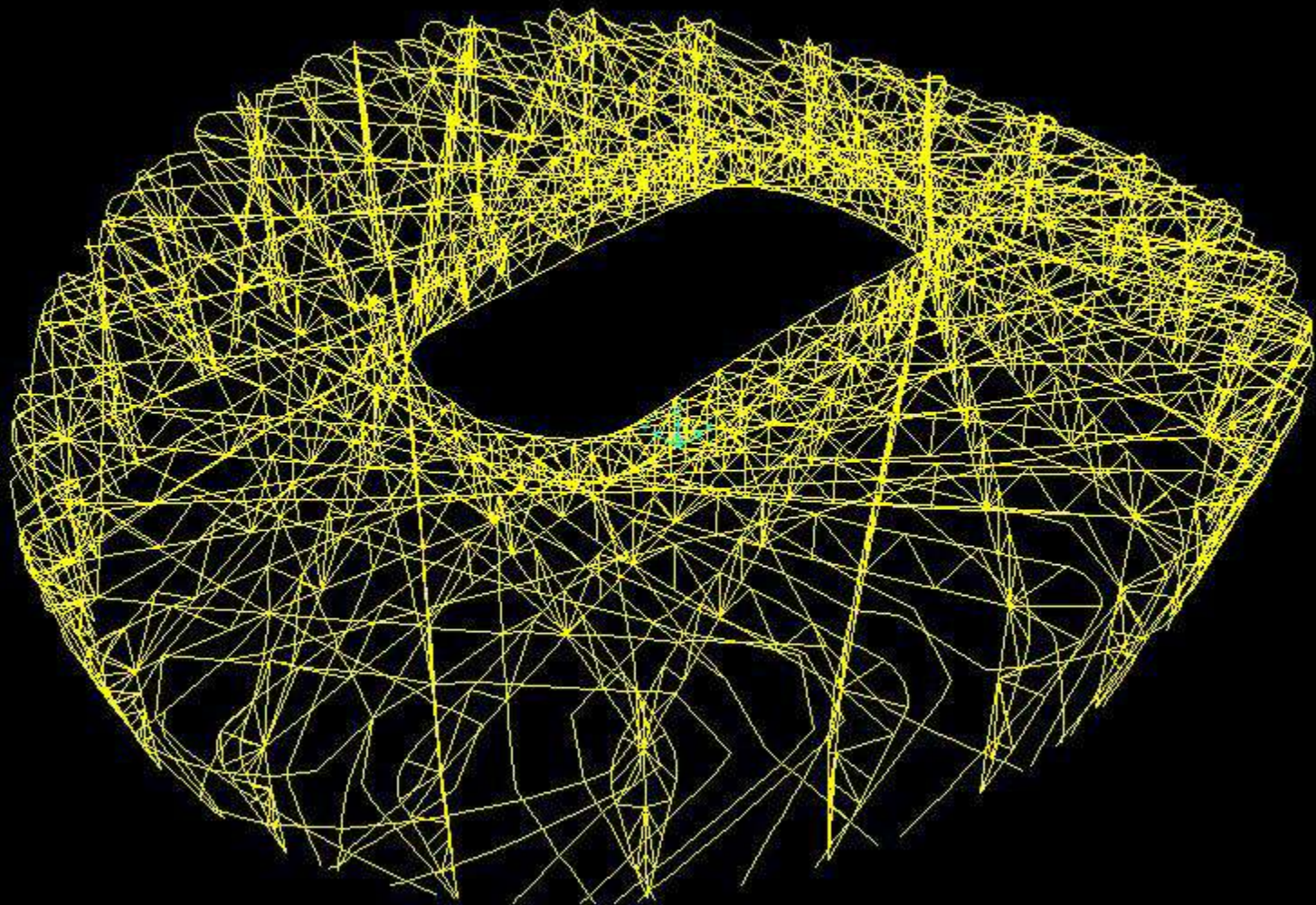


2 3d modelling pattern







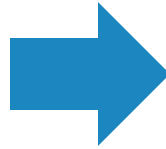


Optimum design of structures:

- 1、Auto section selection 自动选择截面
- 2、Structural check 结构验算
- 3、Report output 计算书生成

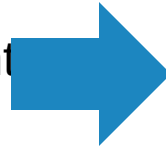
Element 建模 单元

Frame element
框架单元



- 三维框架
- 三维桁架
- 平面框架
- 平面网架
- 平面桁架

Shell element
壳单元



- 三维壳，如罐和穹
- 板结构，如楼板
- 膜结构，如剪力墙

Modelling method

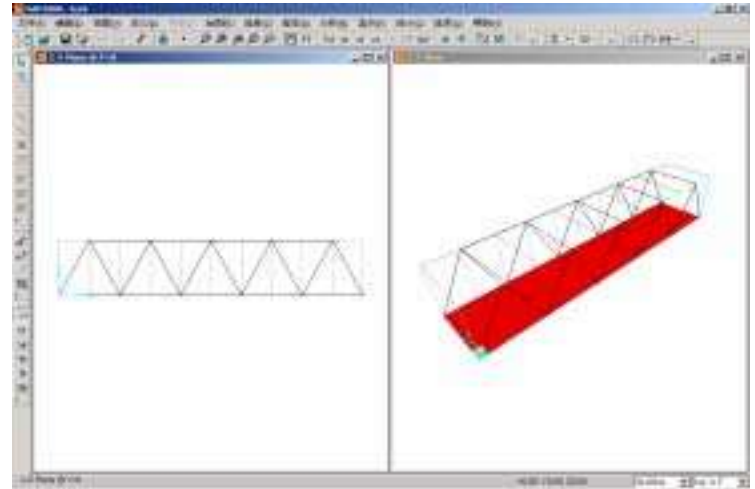
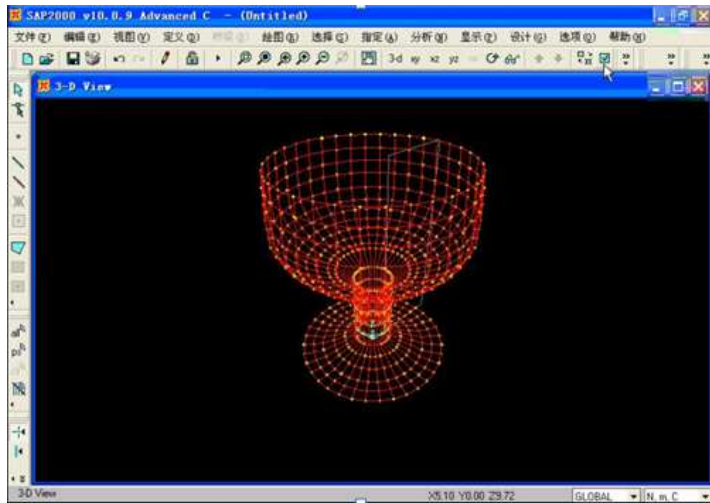
建模
方法

A

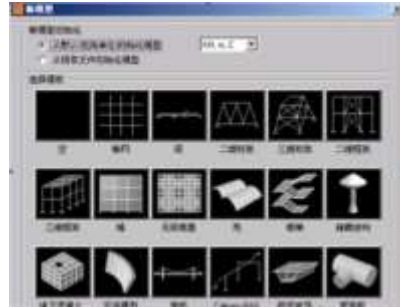
Import model
(CAD, dxf)

B

Default template



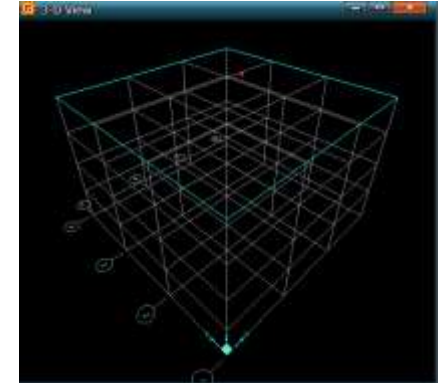
Sketch up



Template



Axial system



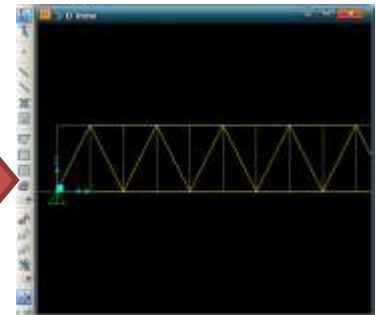
Sketch



Material

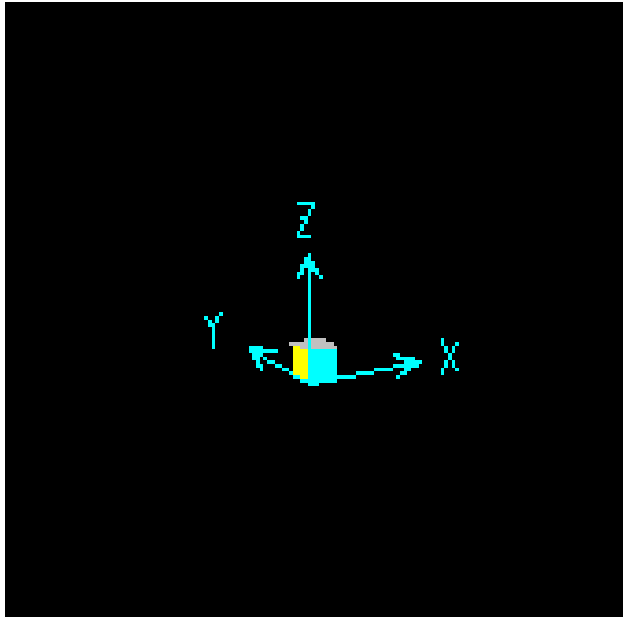


Frame

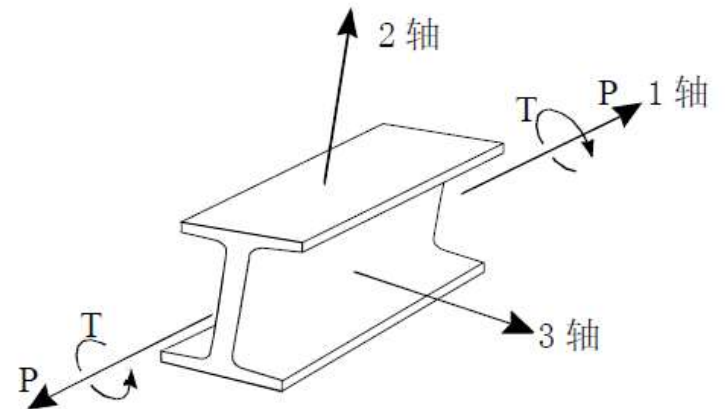


Analysis

Coordinate system



Global coordinate



Local coordinate

Nonlinear analysis 非线性分析:

Pushover analysis: is a nonlinear static calculation for seismic and progressive collapse analysis of structures. The load pattern is inferred from dynamic calculation of equivalent loads. These loads are incrementally increased until the global failure of the structure occurs. The nonlinear behaviour of a building is usually represented by using plastic hinges.

References

- SAP2000 User's manual
- SAP2000 使用指南

Applications

1. Geometric modelling
2. Model simplification
3. Interactions
4. Death and live
5. Cracks
6. Multi-step analysis
7. Analysis & design

THANKS